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## UNIT - 5

## Higher order differential equations

> linear differential equation of higher order with constant co-efficient.

> The general form of  $n^{\text{th}}$  order linear differential equation with constant co-efficient is

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = Q(x) \text{ --- (1)}$$

> where,  $a_0, a_1, a_2, \dots, a_n \neq 0$  are constants and  $Q(x)$  is a function of  $x$  only.

> The right hand side of eqn (1) is called the non-homogenous term.

> If  $Q(x)$  is identically zero in eqn (1), we have

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = 0 \text{ --- (2)}$$

> Hence, equation (2) is called the homogenous linear equation corresponding to the eqn (1).

To solve the homogenous differential equation (2):

> The solution eqn (2) is obtained constructing auxiliary (or) subsidiary equation in the way of replacing

$$\frac{dy}{dx} = m, \quad \frac{d^2 y}{dx^2} = m^2, \quad \dots, \quad \frac{d^n y}{dx^n} = m^n$$

> Then, we have

$$a_n m^n + a_{n-1} m^{n-1} + \dots + a_1 m + a_0 = 0 \text{ --- (3)}$$

> while, solving eqn (2) one of the following 3 cases may occur

- 1) All the roots are real & distinct / different
- 2) All the roots are real but some of the roots are repeating
- 3) All the roots are real & imaginary.

$y = \text{Complementary function} + \text{Particular integral}$

$$y = C.F + P.I$$

$D \rightarrow \text{Differential Operator}$

$$D = \frac{d}{dx}$$

$$D^2 = \frac{d^2}{dx^2}$$

To find C.F :

- 1) Roots are real & distinct : (2 roots  $\rightarrow m_1, m_2$ )

$$C.F = Ae^{m_1 x} + Be^{m_2 x} \quad \text{eg: } m_1 = 1, m_2 = 2$$
$$C.F = Ae^x + Be^{2x}$$

- 2) Roots are real & equal :  $[m_1 = 2, m_2 = 2]$

$$C.F = (A + Bx)e^{mx}$$

(or)

$$C.F = (Ax + B)e^{mx}$$

$$\text{eg: } m_1 = 2, m_2 = 2$$

$$C.F = (A + Bx)e^{2x}$$

$$\text{Eg: } m_1 = 0, 2, 2$$

$$C.F = (A + Bx)e^{2x} + Ce^x$$

- 3) Roots are real & imaginary:

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x) \quad \boxed{m = \alpha \pm i\beta}$$

$$m = 1 \pm i2$$

$$C.F = e^x (A \cos 2x + B \sin 2x)$$

## Operator D :

> A mathematical device by means of which we can convert one function to another is known as Operator.

> Where, the differential operator D converts differential function  $f(x)$  into a new function  $f'(x)$

$$\text{So, } \frac{d}{dx} = D$$

$$\frac{d^2}{dx^2} = D^2$$

$$\frac{d^3}{dx^3} = D^3$$

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + 2 = 0$$

$$D^2 y + Dy + 2 = 0$$

From eqn ①

$$a_n D^n y + a_{n-1} D^{n-1} y + \dots + a_1 Dy + a_0 y = A(x)$$

$$[a_n D^n + a_{n-1} D^{n-1} + \dots + a_1 D + a_0] y = A(x)$$

↓  
P.I

$$A(x) = 0 \rightarrow \text{P.I} = 0$$

### Type-I

Prblms on homogeneous differential equation

1) Solve  $(D^2 + 5D + 6)y = 0$

Soln:

$$\text{Gn } (D^2 + 5D + 6)y = 0$$

The Auxiliary eqn is

[replace D by m]

$$m^2 + 5m + 6 = 0$$

$$m^2 + 2m + 3m + 6 = 0$$

$$m(m+2) + 3(m+2) = 0$$

$$(m+2)(m+3) = 0$$

$$m+2 = 0 \quad | \quad m+3 = 0$$

$$m = -2 \quad | \quad m = -3$$

$$\boxed{m = -2, -3}$$

The roots are real & distinct

$$\begin{array}{c} 6 \\ \wedge \\ 3 \quad 2 \\ \vee \\ 5 \end{array}$$

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{-2x} + Be^{-3x}$$

$$P.I = 0$$

General equation is

$$y = C.F + P.I$$

$$y = Ae^{-2x} + Be^{-3x}$$

2) Solve  $(D^2 + 2D + 1)y = 0$

Soln:

$$\text{Gn. } (D^2 + 2D + 1)y = 0$$

The A.E is

$$m^2 + 2m + 1 = 0$$

$$(m+1)(m+1) = 0$$

$$m+1=0, m+1=0$$

$$m = -1, -1$$

The roots are real & equal

$$C.F = (A+Bx)e^{mx}$$

$$C.F = (A+Bx)e^{-x}$$

$$P.I = 0$$

Gnrl eqn

$$y = C.F + P.I$$

$$y = (A+Bx)e^{-x}$$

3) Solve  $(D^2 + D + 1)y = 0$

Soln

$$\text{Gn } (D^2 + D + 1)y = 0$$

The A.E is

$$m^2 + m + 1 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1, b=1, c=1$$

$$m = \frac{-1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2a}$$

$$= \frac{-1 \pm \sqrt{1-4}}{2a}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

$$m = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$$

The roots are real & imaginary

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$\alpha = -1/2 \quad \beta = \sqrt{3}/2$$

$$C.F = e^{-1/2 x} (A \cos(\sqrt{3}/2 x) + B \sin(\sqrt{3}/2 x))$$

General eqn

$$y = C.F + P.I$$

$$y = e^{-1/2 x} (A \cos(\sqrt{3}/2 x) + B \sin(\sqrt{3}/2 x))$$

4) Solve  $(D^2 - 4D + 5)y = 0$

Soln:

$$\text{Gn: } (D^2 - 4D + 5)y = 0$$

The A.E is

$$(m^2 - 4m + 5) = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1, b=-4, c=5$$

$$m = \frac{4 \pm \sqrt{16 - 20}}{2}$$

$$m = \frac{4 \pm i2}{2}$$

$$m = \frac{4}{2} \pm \frac{2i}{2}$$

$$m = 2 \pm i$$

The roots are real & imaginary

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$C.F = e^{2x} (A \cos x + B \sin x)$$

General eqn

$$y = C.F + P.I$$

$$y = e^{2x} (A \cos x + B \sin x)$$

5) Solve  $\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = 0$

$$D^3 y + 6D^2 y + 11Dy + 6y = 0$$

$$(D^3 + 6D^2 + 11D + 6)y = 0$$

The A.E is

$$m^3 + 6m^2 + 11m + 6 = 0$$

[Using Synthetic division method]

$$\begin{array}{r|rrrr} -1 & 1 & 6 & 11 & 6 \\ & & -1 & -5 & -6 \\ \hline & 1 & 5 & 6 & 0 \end{array}$$

$$m = -1$$

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m+2=0 \quad m+3=0$$

$$m=-2 \quad m=-3$$

∴ The roots are  $m = -1, -2, -3$

∴ The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x} + Ce^{m_3 x}$$

$$C.F = Ae^{-x} + Be^{-2x} + Ce^{-3x}$$

General eqn

$$y = C.F + P.I$$

$$y = Ae^{-x} + Be^{-2x} + Ce^{-3x}$$

b) Solve  $\frac{d^3 y}{dx^3} - 3 \frac{dy}{dx} + 2y = 0$

Soln:

$$D^3 y - 3Dy + 2y = 0$$

$$(D^3 - 3D + 2)y = 0$$

The A.E is

$$m^3 - 3m + 2 = 0$$

[using synthetic division method]

$$\begin{array}{r|rrrrr} 1 & 1 & 0 & -3 & 2 & \\ & & 0 & 1 & 1 & -2 \\ \hline & 1 & 1 & -2 & 0 & \end{array}$$

$$m = 1$$

$$m^2 + m - 2 = 0$$

$$(m-1)(m+2) = 0$$

$$m = 1, -2$$

$$\begin{array}{r|rrrrr} -2 & 1 & 0 & -3 & 2 & \\ & & 0 & -2 & 4 & -2 \\ \hline & 1 & -2 & 1 & 0 & \end{array}$$

$$\begin{array}{r|rrrrr} -1 & 1 & 0 & -3 & 2 & \\ & & 0 & -1 & 1 & 2 \\ \hline & 1 & -1 & -2 & 0 & \end{array}$$

$$\begin{array}{r|rrrrr} 1 & 1 & 0 & -3 & 2 & \\ & & 0 & 1 & 1 & -2 \\ \hline & 1 & 1 & -2 & 0 & \end{array}$$

$$\begin{array}{c} -2 \\ \wedge \\ +2 \quad -1 \end{array}$$

∴ The roots are  $m = 1, 1, -2$

∴ The roots are ~~real & equal & distinct~~

$$C.F = (A+Bx)e^{m_1x} + Ce^{m_2x}$$

$$C.F = (A+Bx)e^x + Ce^{-2x}$$

General eqn

$$y = C.F + P.I$$

$$y = (A+Bx)e^x + Ce^{-2x}$$

7) Solve  $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} - 6\frac{dy}{dx} = 0$

Soln:

$$D^3y - D^2y - 6Dy = 0$$

$$(D^3 - D^2 - 6D)y = 0$$

The A.E is

$$m^3 - m^2 - 6m = 0$$

[Using synthetic division method]

$$+ \quad - \quad - \quad 6 \quad 0$$

$$m(m^2 - m - 6) = 0$$

$$m = 0$$

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

$$m = 3, -2$$

∴ The roots are  $m = 0, 3, -2$

$$\begin{array}{r} -6 \\ \wedge \\ -3 \quad 2 \\ \vee \\ -1 \end{array}$$

∴ The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x} + Ce^{m_3 x}$$

$$C.F = Ae^0 + Be^{-2x} + Ce^{3x}$$

General formula:

$$y = C.F + P.I$$

$$y = Ae^0 + Be^{-2x} + Ce^{3x}$$

$$\boxed{y = A + Be^{-2x} + Ce^{3x}}$$

Particular integral (P.I):



$$P.I = \frac{1}{f(D)} Q(x)$$

$$f(D) \rightarrow (D^2 + D + 1)$$

$$Q(x) \rightarrow e^x$$

↓  
right hand side value

Where  $\frac{1}{f(D)}$  is the inverse operator of  $f(D)$  &  $Q(x)$  takes one of the following cases

1.  $e^{ax}$ , where  $a$  is a constant
2.  $\sin ax$  (or)  $\cos ax$ , where  $a$  is a constant
3.  $x^m$ , where  $m$  is a positive integer
4.  $e^{ax} \cdot V$ , where  $V$  is  $\sin ax$  (or)  $\cos ax$  (or)  $x^m$

Type I:

$$Q(x) = e^{ax}, \text{ where } a \text{ is a constant}$$

$$P.I = \frac{1}{f(D)} Q(x), \text{ Replace } D = a$$

$$P.I = \frac{1}{f(a)} \cdot e^{ax}, \text{ Provided } f(a) \neq 0$$

Suppose  $f(a) = 0$

$$\begin{aligned} P.I &= x \cdot \frac{1}{f'(D)} e^{ax}, \text{ Replace } D = a \\ &= x \frac{1}{f'(a)} e^{ax}, \text{ Provided } f'(a) \neq 0 \end{aligned}$$

Suppose  $f'(a) = 0$

$$P.I = x^2 \cdot \frac{1}{f''(D)} e^{ax}$$

Replace  $D = a$

$$= x^2 \cdot \frac{1}{f''(a)} e^{ax}, \quad f''(a) \neq 0$$

If  $f'(a) = 0$  it continues to  $x^3 \dots$

1) Solve  $(D^2 - 4)y = e^{2x}$

Soln

General Soln is  $y = C.F + P.I$

$$\text{Given: } (D^2 - 4)y = e^{2x}$$

The A.E is

$$m^2 - 4 = 0$$

$$m^2 = 4$$

$$m = \pm \sqrt{4}$$

$$\boxed{m = \pm 2}$$

The roots are  $m = -2, 2$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$\boxed{C.F = Ae^{-2x} + Be^{2x}}$$

$$P.I = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 4} \cdot e^{2x}$$

Replace  $D = a$

$$e^{ax} = e^{2x}$$

$$= \frac{1}{2^2 - 4} \cdot e^{2x}$$

$$= \frac{1}{0}$$

$$P.I = x \cdot \frac{1}{f'(D)} \cdot e^{2x}$$

$$= x \cdot \frac{1}{2D-0} \cdot e^{2x}$$

$$= x \cdot \frac{e^{2x}}{2(2)}$$

$$P.I = \frac{x e^{2x}}{4}$$

$$y = C.F + P.I$$

$$y = A e^{-2x} + B e^{2x} + \frac{x e^{2x}}{4}$$

2. Solve  $(D^2+4)y = e^{4x} + 5$

Soln:

Ans:

$$(D^2+4)y = e^{4x} + 5$$

$$(D^2+4)y = e^{4x} + 5e^{0x}$$

The A.E is

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm \sqrt{-4}$$

$$m = \pm 2i$$

The roots are  $m = -2i, 2i$

$\therefore$  The roots are imaginary

$$C.F = e^{0x} (A \cos 2x + B \sin 2x)$$

$$= e^{0x} (A \cos 2x + B \sin 2x)$$

$$C.F = (A \cos 2x + B \sin 2x)$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2+4} \cdot e^{4x}$$

Replace  $D = a$

$$= \frac{1}{4^2+4} \cdot e^{4x}$$

$$P.I_1 = \frac{1}{20} e^{4x}$$

$$P.I_2 = \frac{1}{D^2+4} \cdot 5e^{0x}$$

Replace  $D = a$

$$= \frac{1}{0^2+4} \cdot 5e^{0x}$$

$$= \frac{5}{4} = \frac{5}{4} //$$

$$P.I_2 = \frac{5}{4}$$

$$y = C.F + P.I$$

$$y = (A \cos 2x + B \sin 2x) + \frac{1}{20} e^{4x} + \frac{5}{4}$$

3) Solve  $(D^2+5D+6)y = e^x + e^{-2x}$

Soln:

$$\text{Gn: } (D^2+5D+6)y = e^x + e^{-2x}$$

The A.E is

$$m^2+5m+6=0$$

$$(m+3)(m+2)=0$$

$$m+3=0, m+2=0$$

$$m=-3, -2$$

The roots are  $m=-3, -2$

The roots are real & distinct

$$C.F = Ae^{m_1x} + Be^{m_2x}$$

$$C.F = Ae^{-2x} + Be^{-3x}$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2+5D+6} \cdot e^x$$

Replace  $D = a$

$$= \frac{1}{1^2+5(1)+6} \cdot e^x$$

$$P.I_1 = \frac{1}{12} \cdot e^x$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2+5D+6} \cdot e^{-2x}$$

Replace  $D=a$

$$= \frac{1}{(-2)^2+5(-2)+6} \cdot e^{-2x}$$

$$= \frac{1}{4-10+6} \cdot e^{-2x}$$

$$= \frac{1}{0}$$

$$P.I_2 = x \cdot \frac{1}{f'(D)} \cdot e^{-2x} Q(x)$$

$$= x \cdot \frac{1}{2D+5} \cdot e^{-2x}$$

Replace  $D=a$

$$= x \cdot \frac{1}{2(-2)+5} \cdot e^{-2x}$$

$$= x \cdot \frac{1}{1} \cdot e^{-2x}$$

$$\boxed{P.I_2 = x \cdot e^{-2x}}$$

$$y = C.F + P.I$$

$$\boxed{y = Ae^{-2x} + Be^{-3x} + \frac{1}{12} \cdot e^x + x \cdot e^{-2x}}$$

4. Solve  $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3e^{-2x} + e^{-3x}$

Soln:

$$\text{Ans: } \frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3e^{-2x} + e^{-3x}$$

$$D^2y + 5Dy + 6y = 3e^{-2x} + e^{-3x}$$

$$(D^2+5D+6)y = 3e^{-2x} + e^{-3x}$$

The A.E is

$$m^2+5m+6=0$$

$$(m+3)(m+2)=0$$

$$m+3=0, \quad m+2=0$$

$$m = -2, -2$$

The roots are real & distinct

$$C.I. = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.I. = Ae^{-2x} + Be^{-3x}$$

$$P.I._1 = \frac{1}{f(D)} \cdot A(x)$$

$$= \frac{1}{D^2 + 5D + 6} \cdot 3e^{-2x}$$

Replace  $D = a$

$$= \frac{1}{(-2)^2 + 5(-2) + 6} \cdot 3e^{-2x}$$

$$= \frac{1}{4 - 10 + 6} \cdot 3e^{-2x}$$

$$= \frac{1}{0}$$

$$P.I._1 = x \cdot \frac{1}{f'(D)} \cdot A(x)$$

$$= x \cdot \frac{1}{2D + 5} \cdot 3e^{-2x}$$

Replace  $D = a$

$$= x \cdot \frac{1}{2(-2) + 5} \cdot 3e^{-2x}$$

$$P.I._1 = x \cdot 3e^{-2x}$$

$$P.I._2 = \frac{1}{f(D)} \cdot A(x)$$

$$= \frac{1}{D^2 + 5D + 6} \cdot e^{-3x}$$

Replaced  $D = a$

$$= \frac{1}{(-3)^2 + 5(-3) + 6} \cdot e^{-3x}$$

$$= \frac{1}{9 - 15 + 6} \cdot e^{-3x}$$

$$P \cdot I_2 = \frac{1}{-1} \cdot e^{-3x}$$

$$P \cdot I_2 = x^3 \cdot \frac{1}{f'(D)} \cdot e^{-3x}$$

$$= x \cdot \frac{1}{2D+5} \cdot e^{-3x}$$

$$= x \cdot \frac{1}{2(-3)+5} \cdot e^{-3x}$$

$$= x \cdot \frac{1}{-1} \cdot e^{-3x}$$

$$P \cdot I_2 = x \cdot -e^{-3x}$$

$$P \cdot I_2 = -x e^{-3x}$$

$$y = C \cdot F + P \cdot I$$

$$y = A e^{-2x} + B e^{-3x} + 3x e^{-2x} - x e^{-3x}$$

5) Solve  $(D^2 - 4D + 4)y = \cosh 2x + 3$

~~Ans~~

Ans:

$$(D^2 - 4D + 4)y = \cosh 2x + 3$$

$$= \frac{e^{2x}}{2} + \frac{e^{-2x}}{2} + 3e^{0x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

The A.E is

$$m^2 - 4m + 4 = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2$$

$$m = 2, 2$$

The roots are real & equal

$$C \cdot F = (A + Bx)e^{mx}$$

$$C \cdot F = (A + Bx)e^{2x}$$

$$P \cdot I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 4D + 4} \cdot \frac{e^{2x}}{2}$$

Replace  $D = a$

$$= \frac{1}{2^2 - 4(2) + 4} \cdot \frac{e^{2x}}{2}$$

$$= \frac{1}{4 - 8 + 4} \cdot \frac{e^{2x}}{2}$$

$$= \frac{1}{0}$$

$$P.I_1 = x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{2D - 4} \cdot Q(x)$$

Replace  $D = a$

$$= x \cdot \frac{1}{2(2) - 4} \cdot \frac{e^{2x}}{2}$$

$$= x \cdot \frac{1}{0}$$

$$P.I_1 = x^2 \cdot \frac{1}{f''(D)} \cdot \frac{e^{2x}}{2}$$

$$= x^2 \cdot \frac{1}{2} \cdot \frac{e^{2x}}{2}$$

$$\boxed{P.I_1 = \frac{x^2 \cdot e^{2x}}{4}}$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 4D + 4} \cdot \frac{e^{-2x}}{2}$$

Replace  $D = a$

$$= \frac{1}{(-2)^2 - 4(-2) + 4} \cdot \frac{e^{-2x}}{2}$$

$$= \frac{1}{4 + 8 + 4} \cdot \frac{e^{-2x}}{2}$$

$$= \frac{1}{16} \cdot \frac{e^{-2x}}{2}$$

$$\boxed{P.I_2 = \frac{e^{-2x}}{32}}$$

$$P.I_3 = \frac{1}{f(D)} Q(x)$$

$$= \frac{1}{D^2 - 4D + 4} \cdot 3e^{0x}$$

Replace  $D = a$

$$= \frac{1}{4} \cdot 3$$

$$\boxed{P.I_3 = \frac{3}{4}}$$

$$y = C.F + P.I$$

$$\boxed{y = (A + Bx)e^{2x} + \frac{x^2 e^{2x}}{4} + \frac{e^{-2x}}{32} + \frac{3}{4}}$$

⑨ Solve  $(D^2 - 9)y = \sinh 3x$

Soln:  
Ans

$$\begin{aligned} (D^2 - 9)y &= \sinh 3x \\ &= \frac{e^{3x}}{2} - \frac{e^{-3x}}{2} \end{aligned}$$

The A.E is

$$m^2 - 9 = 0$$

$$m^2 = 9$$

$$m = \pm \sqrt{9}$$

$$\boxed{m = \pm 3}$$

The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$\boxed{C.F = Ae^{-3x} + Be^{3x}}$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 9} \cdot \frac{e^{3x}}{2}$$

Replace  $D = a$

$$= \frac{1}{3^2 - 9} \cdot \frac{e^{3x}}{2}$$

$$= \frac{1}{6}$$

$$P.I_1 = x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{2D} \cdot \frac{e^{3x}}{2}$$

$$= x \cdot \frac{1}{6} \cdot \frac{e^{3x}}{2}$$

$$P.I_1 = \frac{x \cdot e^{3x}}{12}$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 9} \cdot \frac{e^{-3x}}{2}$$

$$= \frac{1}{(-3)^2 - 9} \cdot \frac{e^{-3x}}{2}$$

$$= \frac{1}{6}$$

$$P.I_2 = x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{2D} \cdot \frac{e^{-3x}}{2}$$

$$= x \cdot \frac{1}{2(-3)} \cdot \frac{e^{-3x}}{2}$$

$$P.I_2 = \frac{x \cdot e^{-3x}}{-12}$$

$$y = Ae^{-3x} + Be^{3x} + \frac{x \cdot e^{3x}}{12} + \frac{x \cdot e^{-3x}}{12}$$

$$y = Ae^{-3x} + Be^{3x} + \frac{x}{12} [e^{3x} + e^{-3x}]$$

$$y = Ae^{-3x} + Be^{3x} + \frac{x}{12} 2 \cosh 3x$$

$$y = Ae^{-3x} + Be^{3x} + \frac{x}{6} \cosh 3x$$

Type II:

If  $Q(x) = \cos ax$  (or)  $\sin ax$ , where  $a$  is a constant

$$P.I = \frac{1}{f(D^2)} \sin ax, \text{ Replace } D^2 = -a^2$$

$$= \frac{1}{f(-a^2)} \sin ax, \text{ if } f(-a^2) \neq 0$$

$$\frac{1}{D} = \int$$

$$\frac{1}{D^2} = \int \int$$

$$\text{Ex: } \frac{1}{D^2+4} \cos 2x \text{ or } \frac{1}{D^2+4} \sin 2x$$

$$\frac{1}{D^2+4} \cos 2x$$

1) Solve  $(D^2+4)y = \cos 2x$

Soln

Ans:

$$(D^2+4)y = \cos 2x$$

The A.E is

$$m^2+4=0$$

$$m^2=-4$$

$$m = \pm \sqrt{-4}$$

$$m = \pm 2i$$

$$m = 2i, -2i$$

$\therefore$  The roots are imaginary

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$= e^{0x} (A \cos 2x + B \sin 2x)$$

$$C.F = A \cos 2x + B \sin 2x$$

$$P.I = \frac{1}{f(D^2)} \cos ax$$

$$P.I = \frac{1}{D^2+4} \cos 2x$$

$$= \frac{1}{-4+4} \cos 2x$$

$$= \frac{1}{0}$$

$$\text{Replace } D^2 = -a^2$$

$$= -(2^2)$$

$$= -4$$

$$P.I = x \cdot \frac{1}{2D} \cdot \cos 2x$$

$$= \frac{x}{2} \cdot \frac{1}{D} \cos 2x$$

$$= \frac{x}{2} \int \cos 2x \, dx$$

$$= \frac{x}{2} \cdot \frac{\sin 2x}{2}$$

$$P.I = \frac{x \sin 2x}{4}$$

$$y = C.F + P.I$$

$$y = A \cos 2x + B \sin 2x + \frac{x \sin 2x}{4}$$

2) Solve  $(D^2 - 4D + 4)y = \sin 2x$

Soln

Ans:

$$(D^2 - 4D + 4)y = \sin 2x$$

The A.E is

$$m^2 - 4m + 4 = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2$$

∴ The roots are real & equal

$$C.F = (A + Bx)e^{mx}$$

$$C.F = (A + Bx)e^{2x}$$

$$P.I = \frac{1}{f(D^2)} \cdot \sin ax$$

$$= \frac{1}{D^2 - 4D + 4} \cdot \sin 2x$$

diff

$$\sin = \cos$$

$$\cos = -\sin$$

∫

$$\sin = -\cos$$

$$\cos = \sin$$

Replace  $D^2 = -a^2$

$$\begin{aligned}
 &= \frac{1}{-4 - 4D + 4} \cdot \sin 2x \\
 &= \frac{1}{-4} \cdot \frac{1}{D} \sin 2x \\
 &= \frac{1}{-4} \int \sin 2x \, dx \\
 &= \frac{1}{-4} \cdot \frac{-\cos 2x}{2}
 \end{aligned}$$

$$P.I = \frac{\cos 2x}{8}$$

$$y = C.F + P.I$$

$$y = (A + Bx)e^{2x} + \frac{\cos 2x}{8}$$

3) Solve  $(D^2 + 5D + 6)y = \cos^2 x$

Soln:  
Gn:

$$(D^2 + 5D + 6)y = \cos^2 x$$

$$= \frac{1 + \cos 2x}{2}$$

$$= \frac{1}{2} + \frac{\cos 2x}{2}$$

$$= \frac{1}{2} e^{0x} + \frac{1}{2} \cos 2x$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

The A.E is

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m = -2 \quad m = -3$$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{-2x} + Be^{-3x}$$

$$P.I_1 = \frac{1}{f(D)} \cdot \frac{1}{2} e^{0x}$$

$$P.I_1 = \frac{1}{D^2 + 5D + 6} \cdot \frac{1}{2} e^{0x}$$

Replace  $D = a$

$$= \frac{1}{0 + 5(0) + 6} \cdot \frac{1}{2} e^{0x}$$

$$P.I_1 = \frac{1}{12}$$

$$P.I_2 = \frac{1}{f(D)} \cdot G(x)$$

$$= \frac{1}{D^2 + 5D + 6} \cdot \frac{1}{2} \cos 2x$$

Replace  $D^2 = -(a^2)$

$$= -(2^2)$$

$$= -4$$

$$= \frac{1}{-4 + 5D + 6} \cdot \frac{1}{2} \cos 2x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D + 2)} \cdot \cos 2x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D + 2)} \times \frac{(5D - 2)}{(5D - 2)} \cdot \cos 2x$$

$$= \frac{1}{2} \cdot \frac{5D - 2}{(5D^2 - 2^2)} \cdot \cos 2x$$

$$= \frac{1}{2} \cdot \frac{5D - 2}{25(-4) - 4} \cdot \cos 2x$$

$$= \frac{1}{2} \cdot \frac{5D - 2}{-104} \cdot \cos 2x$$

$$= \frac{1}{2} \cdot \frac{5D(\cos 2x) - 2 \cos 2x}{[25(-4) - 4]}$$

$D(\cos 2x)$

$$\frac{d}{dx} (\cos 2x) = -\sin 2x$$

$$= \frac{1}{2} \cdot \frac{[5(-2 \sin 2x) - 2 \cos 2x]}{-104}$$

$$= \frac{1}{2} \cdot \frac{[-5 \sin 2x - \cos 2x]}{-104}$$

$$P.I_2 = \frac{5 \sin 2x + \cos 2x}{104}$$

$$y = C.F + P.I$$

$$y = Ae^{-2x} + Be^{-3x} + \frac{1}{12} + \frac{5 \sin 2x + \cos 2x}{104}$$

A Solve  $(D^2+1)y = \sin^3 x$

Soln  
Gn:

$$\sin^3 \theta = \frac{1}{4} [3 \sin \theta - \sin 3\theta]$$

$$\begin{aligned} (D^2+1)y &= \sin^3 x \\ &= \frac{1}{4} [3 \sin x - \sin 3x] \\ &= \frac{3 \sin x}{4} - \frac{\sin 3x}{4} \end{aligned}$$

The A.E is

$$\begin{aligned} m^2+1 &= 0 \\ m^2 &= -1 \\ m &= \pm \sqrt{-1} \\ m &= \pm i \end{aligned}$$

$\therefore$  The roots are imaginary

$$\begin{aligned} C.F &= e^{\alpha x} (A \cos \beta x + B \sin \beta x) \\ &= e^{0x} (A \cos x + B \sin x) \end{aligned}$$

$$C.F = A \cos x + B \sin x$$

$$P.I_1 = \frac{1}{f(D^2)} \cdot \frac{3}{4} \sin x$$

$$= \frac{1}{D^2+1} \cdot \frac{3}{4} \sin x$$

$$= \frac{1}{-1+1} \cdot \frac{3}{4} \sin x$$

$$= \frac{1}{0}$$

$$\begin{aligned} \text{Replace } D^2 &= -(a^2) \\ &= -(1^2) \\ &= -1 \end{aligned}$$

$$P.I_1 = x \cdot \frac{1}{f'(D^2)} \cdot \sin x$$

$$= x \cdot \frac{1}{2D+1} \cdot \frac{3}{4} \sin x$$

$$= x \cdot \frac{3}{8} \cdot \frac{1}{D} \sin x$$

$$= x \cdot \frac{3}{8} \int \sin x \, dx$$

$$= \frac{3x}{8} \cdot -\cos x$$

$$P.I_1 = -\frac{\cos x \cdot 3x}{8}$$

$$P.I_2 = \frac{1}{f(D^2)} \cdot \sin ax$$

$$= \frac{1}{D^2+1} \cdot \frac{1}{4} \sin 3x$$

Replace  $D^2 = -a^2$

$$D^2 = -(3^2)$$

$$= -9$$

$$= \frac{1}{-8} \cdot \frac{1}{4} \sin 3x$$

$$P.I_2 = -\frac{\sin 3x}{32}$$

$$y = C.F + P.I$$

$$y = A \cos x + B \sin x - \frac{\cos x \cdot 3x}{8} + \frac{\sin 3x}{32}$$

5. Solve  $(D^2 + 5D + 6)y = \sin 3x \cos x$

Soln:

bn

$$(D^2 + 5D + 6)y = \sin 3x \cos x$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$= \frac{1}{2} [\sin 4x + \sin 2x]$$

$$= \frac{1}{2} \sin 4x + \frac{1}{2} \sin 2x$$

The A.E is

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m = -2, -3$$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{-2x} + Be^{-3x}$$

$$P.I_1 = \frac{1}{f(D^2)} \cdot \sin ax$$

$$= \frac{1}{D^2 + 5D + 6} \cdot \frac{1}{2} \sin 4x$$

$$\text{Replace } D^2 = -(a^2)$$

$$= -(4^2)$$

$$= -16$$

$$= \frac{1}{-16 + 5D + 6} \cdot \frac{1}{2} \sin 4x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D - 10)} \cdot \sin 4x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D - 10)} \times \frac{(5D + 10)}{(5D + 10)} \sin 4x$$

$$= \frac{1}{2} \cdot \frac{(5D + 10)}{(5D)^2 - 10^2} \cdot \sin 4x$$

$$= \frac{1}{2} \cdot \frac{(5D + 10) \sin 4x}{25D^2 - 100}$$

$$= \frac{1}{2} \cdot \frac{(5D + 10) \sin 4x}{25(-16) - 100}$$

$$\text{Replace } D^2 = -(a^2)$$

$$= \frac{1}{2} \cdot \frac{(5(4 \cos 4x) + 10 \sin 4x)}{-500}$$

$$= \frac{10}{2} \cdot \frac{2 \cos 4x + \sin 4x}{-500}$$

$$P.I_1 = \frac{2 \cos 4x + \sin 4x}{-100}$$

$$P.I_2 = \frac{1}{f(D^2)} \sin ax$$

$$= \frac{1}{D^2 + 5D + 6} \cdot \frac{1}{2} \sin 2x$$

$$\text{Rep } D^2 = -(a^2)$$

$$= -(2^2)$$

$$= -4$$

$$= \frac{1}{2} \cdot \frac{1}{(-4) + 5D + 6} \cdot \sin 2x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D + 2)} \cdot \sin 2x$$

$$= \frac{1}{2} \cdot \frac{1}{(5D + 2)} \times \frac{(5D - 2)}{(5D - 2)} \cdot \sin 2x$$

$$= \frac{1}{2} \frac{(5D-2) \sin 2x}{(5D)^2 - 2^2}$$

$$= \frac{1}{2} \frac{(5D \sin 2x - 2 \sin 2x)}{25(-4) - 4}$$

$$= \frac{1}{2} \frac{(5(2 \cos 2x) - 2 \sin 2x)}{-104}$$

$$= \frac{2}{2} \frac{(5 \cos 2x - \sin 2x)}{-104}$$

$$P.I_2 = - \left( \frac{5 \cos 2x - \sin 2x}{104} \right)$$

$$y = C.F + P.I$$

$$y = Ae^{-2x} + Be^{-3x} - \frac{2 \cos 4x + \sin 4x}{100} - \frac{5 \cos 2x - \sin 2x}{104}$$

6. Solve  $(D^2 - 3D + 2)y = \cos 3x \cdot \cos 2x$

Soln:

Ans:

$$\cos a \cos b = \frac{1}{2} (\cos(a+b) + \cos(a-b))$$

$$(D^2 - 3D + 2)y = \cos 3x \cdot \cos 2x$$

$$= \frac{1}{2} (\cos 5x + \cos x)$$

$$= \frac{1}{2} \cos 5x + \frac{1}{2} \cos x$$

The A.E is

$$m^2 - 3m + 2 = 0$$

$$(m-2)(m-1) = 0$$

$$m = 2, 1$$

$$\begin{array}{c} 2 \\ \wedge \\ -2 \quad -1 \\ \vee \\ -3 \end{array}$$

∴ The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{2x} + Be^x$$

$$P.I_1 = \frac{1}{f(D^2)} \cdot \cos ax$$

$$= \frac{1}{D^2 - 3D + 2} \cdot \frac{1}{2} \cos 5x$$

$$= \frac{1}{-25 - 3D + 2} \cdot \frac{1}{2} \cos 5x$$

$$= \frac{1}{-3D - 23} \cdot \frac{1}{2} \cos 5x$$

$$= \frac{1}{2} \cdot \frac{1}{-3D - 23} \times \frac{-3D + 23}{-3D + 23} \cdot \cos 5x$$

$$= \frac{1}{2} \cdot \frac{-3D + 23 \cdot \cos 5x}{(-3D)^2 - 23^2}$$

$$= \frac{1}{2} \cdot \frac{-3D + 23 \cdot \cos 5x}{9(-25) - 529}$$

$$= \frac{1}{2} \cdot \frac{+3D \cos 5x + 23 \cos 5x}{+754}$$

$$= \frac{1}{2} \cdot \frac{3 \cdot (-5 \sin 5x) + 23 \cos 5x}{754}$$

$$= \frac{-15 \sin 5x + 23 \cos 5x}{1508}$$

$$P.I_1 = - \frac{15 \sin 5x + 23 \cos 5x}{1508}$$

$$P.I_2 = \frac{1}{f(D^2)} \cdot \cos ax$$

$$= \frac{1}{D^2 - 3D + 2} \cdot \frac{1}{2} \cos x$$

$$= \frac{1}{2} \cdot \frac{1}{-1 - 3D + 2} \cdot \cos x$$

$$= \frac{1}{2} \cdot \frac{1}{-3D + 1} \cdot \cos x$$

$$= \frac{1}{2} \cdot \frac{1}{-3D + 1} \times \frac{-3D - 1}{-3D - 1} \cdot \cos x$$

$$\text{Rep } D^2 = -(a^2)$$

$$= -(5^2)$$

$$= -(25)$$

$$= -25$$

$$\text{Rep } D^2 = -(a^2)$$

$$= -(1^2)$$

$$= -1$$

$$= \frac{1}{2} \cdot \frac{(-3D-1) \cdot \cos x}{(-3D)^2 - 1^2}$$

$$= \frac{1}{2} \cdot \frac{(-3D-1) \cos x}{(9(-1) - 1)}$$

$$= \frac{1}{2} \cdot \frac{-3D \cos x - 1 \cos x}{-10}$$

$$= \frac{1}{2} \cdot \frac{-(3D \cos x + \cos x)}{-10}$$

$$P.I_2 = \frac{-3 \sin x + \cos x}{20}$$

$$y = C.F + P.I$$

$$y = Ae^{2x} + Be^x - \frac{15 \sin 5x + 23 \cos 5x}{1508} - \frac{3 \sin x + \cos x}{20}$$

Type - III

If  $G(x) = x^n$ , where  $n$  is positive integer

$$P.I = \frac{1}{f(D)} \cdot x^n$$

Binomial Series:

$$1) (1-x)^{-1} = 1+x+x^2+x^3+\dots$$

$$2) (1+x)^{-1} = 1-x+x^2-x^3+\dots$$

$$3) (1-x)^{-2} = 1+2x+3x^2+4x^3+\dots$$

$$4) (1+x)^{-2} = 1-2x+3x^2-4x^3+\dots$$

1) Solve  $(D^2 + 5D + 6)y = x$

Soln:

Ans:  $(D^2 + 5D + 6)y = x$

The A.E is

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m = -2, -3$$

∴ The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{-2x} + Be^{-3x}$$

$$P.I = \frac{1}{(D^2 + 5D + 6)} \cdot x$$

$$= \frac{1}{6 \left[ 1 + \frac{D^2 + 5D}{6} \right]} \cdot x$$

$$= \frac{1}{6} \cdot \left[ 1 + \left( \frac{D^2 + 5D}{6} \right) \right]^{-1} \cdot x$$

$$= \frac{1}{6} \cdot \left[ 1 - \left( \frac{D^2 + 5D}{6} \right) + \left( \frac{D^2 + 5D}{6} \right)^2 - \dots \right] \cdot x$$

$$= \frac{1}{6} \cdot \left[ 1 - \frac{D^2}{6} - \frac{5D}{6} + \frac{D^4 + 25D^2 + 10D^3}{36} - \dots \right] \cdot x$$

$$= \frac{1}{6} \cdot \left[ 1 - \frac{D^2}{6} - \frac{5D}{6} + \frac{D^4}{36} + \frac{25D^2}{36} + \frac{10D^3}{36} - \dots \right] \cdot x$$

$$= \frac{1}{6} \cdot \left[ x - \frac{D^2}{6} x - \frac{5D}{6} x + \frac{D^4}{36} x + \frac{25D^2}{36} x + \frac{10D^3}{36} x - \dots \right]$$

$$= \frac{1}{6} [x - 0 - \frac{5}{6}(1) + 0 + 0 + 0]$$

$$P.I = \frac{1}{6} \left[ x - \frac{5}{6} \right]$$

$$D^2(x) = 0$$

$$\frac{dy}{dx}(x) = 1$$

$$\frac{d^2}{dx^2}(x) = 0$$

$$y = C.F + P.I$$

$$y = Ae^{2x} + Be^{3x} + \frac{1}{b} \left[ x - \frac{x}{b} \right]$$

2) Solve  $(D^2 - D)y = x^2$

$$\text{Sol} \quad \text{Giv} \quad (D^2 - D)y = x^2$$

The A.I is

$$m^2 - m = 0$$

$$m(m-1) = 0$$

$$m = 0, +1$$

The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{0x} + Be^{1x}$$

$$\boxed{C.F = A + Be^x}$$

$$P.I = \frac{1}{f(D)} \cdot x^n$$

$$= \frac{1}{D^2 - D} \cdot x^2$$

$$= \frac{1}{-D(1-D)} \cdot x^2$$

$$= -\frac{1}{D} (1-D)^{-1} \cdot x^2$$

$$= -\frac{1}{D} [1 + D + D^2 + D^3 + \dots] x^2$$

$$= -\frac{1}{D} [x^2 + D(x^2) + D^2(x^2) + D^3(x^2) + \dots]$$

$$= -\frac{1}{D} [x^2 + 2x + 2]$$

$$= - \left[ \frac{x^3}{3} + x^2 + 2x \right]$$

$$= - \left[ \frac{x^2}{D} + \frac{D(x^2)}{D} + \frac{D^2(x^2)}{D} + \frac{D^3(x^2)}{D} + \dots \right]$$

$$= - \left[ \frac{1}{D}(x^2) + x^2 + D(x^2) + D^2(x^2) + \dots \right]$$

$$\boxed{P.I = - \left[ \frac{x^3}{3} + x^2 + 2x + 2 \right]}$$

$$y = C.F + P.I$$

$$\boxed{y = A + Be^x - \frac{x^3}{3} - x^2 - 2x - 2}$$

3. Solve  $(D^2 - D + 1)y = (1 - 3x^2 + x^3)$

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$$(D^2 - D + 1)y = (1 - 3x^2 + x^3)$$

The A.E is

$$m^2 - m + 1 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1, b = -1, c = 1$$

$$m = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)}$$

$$m = \frac{1 \pm \sqrt{1-4}}{2}$$

$$m = \frac{1 \pm \sqrt{3}}{2}$$

$$m = \frac{1 \pm \sqrt{3}i}{2}$$

∴ The roots are real & imaginary

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$C.F = e^{\frac{1}{2}x} (A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x)$$

$$P.I = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - D + 1} \cdot (1 - 3x^2 + x^3)$$

$$= \frac{1}{1 - (D - D^2)} \cdot (1 - 3x^2 + x^3)$$

$$= (1 - (D - D^2))^{-1} \cdot (1 - 3x^2 + x^3)$$

$$= [1 + (D - D^2) + (D - D^2)^2 + \dots] \cdot (1 - 3x^2 + x^3)$$

$$= [1 + (D - D^2) + D^2 - 2D^3 + D^4 \dots] \cdot (1 - 3x^2 + x^3)$$

$$= [1 - 3x^2 + x^3 + (D - D^2)(1 - 3x^2 + x^3) + (D^2 - 2D^3 + D^4)(1 - 3x^2 + x^3) + \dots]$$

$$= [1 - 3x^2 + x^3 + D(1 - 3x^2 + x^3) - D^2(1 - 3x^2 + x^3) + D^2(1 - 3x^2 + x^3) - 2D^3(1 - 3x^2 + x^3) + D^4(1 - 3x^2 + x^3) + \dots]$$

$$= [1 - 3x^2 + x^3 + 0 - 6x + 3x^2 - 0 + 6 - 6x + 0 - 6 + 6x - 2(0 - 0 + 6) + (0 - 0 + 0)]$$

$$= 1 + x^3 - 6x - 12$$

$$P.I = x^3 - 6x - 11$$

$$y = C.F + P.I$$

$$y = e^{\frac{x}{2}} (A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x) + x^3 - 6x - 11$$

4. Solve  $(D^4 - 2D^3 + D^2)y = x^2 + e^x$

Ans:  $(D^4 - 2D^3 + D^2)y = x^2 + e^x$

The A.E is

$$m^4 - 2m^3 + m^2 = 0$$

using  $\star$   $m^2(m^2 - 2m + 1) = 0$

$$m^2 = 0, m^2 - 2m + 1 = 0$$

$$m = 0, 0$$

$$m^2 - 2m + 1 = 0$$

$$(m-1)(m-1) = 0$$

$$\boxed{m = 1, 1}$$

$\therefore$  The roots are 0, 0, 1, 1

$\therefore$  The roots are real & equal

$$C.F = (Ax+B)e^{mx} + (Cx+D)e^{mx}$$

$$C.F = (Ax+B)e^{0x} + (Cx+D)e^x$$

$$\boxed{C.F = (Ax+B) + (Cx+D)e^x}$$

$$P.I, = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^4 - 2D^3 + D^2} \cdot x^2$$

$$= \frac{1}{D^2} \cdot \frac{1}{D^2 - 2D + 1} \cdot x^2$$

$$= \frac{1}{D^2} \cdot \frac{1}{(1-D)^2} \cdot x^2$$

$$= \frac{1}{D^2} \cdot [1-D]^2 \cdot x^2$$

$$= \frac{1}{D^2} \cdot [1+2D+3D^2+4D^3+5D^4] \cdot x^2$$

$$= \left[ \frac{1}{D^2} + \frac{2D}{D^2} + \frac{3D^2}{D^2} + \frac{4D^3}{D^2} + \frac{5D^4}{D^2} \right] x^2$$

$$= \left[ \frac{1}{D^2} (x^2) + \frac{2D}{D^2} (x^2) + \frac{3D^2}{D^2} (x^2) + \frac{4D^3}{D^2} (x^2) + \frac{5D^4}{D^2} (x^2) \right]$$

$$= \left[ \frac{x^4}{12} + 2 \frac{x^3}{3} + 3x^2 + 8x + 10 \right]$$

$$P.I_1 = \frac{x^4}{12} + \frac{2}{3} x^3 + 3x^2 + 8x + 10$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^4 - 2D^3 + D^2} \cdot e^x$$

Replace  $D = a$   
 $D = 1$

$$= \frac{1}{(1)^4 - 2(1)^3 + 1^2} \cdot e^x$$

$$= \frac{1}{1-2+1} \cdot e^x$$

$$= \frac{1}{2-2} \cdot e^x = \frac{1}{0} \cdot e^x$$

$$P.I_2 = x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{4D^3 - 2(3D^2) + 2D} \cdot e^x$$

$$= x \cdot \frac{1}{4(1)^3 - 2(3(1)^2) + 2(1)} \cdot e^x$$

$$= x \cdot \frac{1}{4-6+2} \cdot e^x$$

$$= x \cdot \frac{1}{0} \cdot e^x$$

$$P.I_2 = x^2 \cdot \frac{1}{f''(D)} \cdot Q(x)$$

$$= x^2 \cdot \frac{1}{12D^2 - 12D + 2} \cdot e^x$$

$$= x^2 \cdot \frac{1}{12(1^2) - 12(1) + 2} \cdot e^x$$

$$= x^2 \cdot \frac{1}{12 - 12 + 2} \cdot e^x$$

$$P.I_2 = \frac{x^2 e^x}{2}$$

$$y = C.F + P.I$$

$$y = (Ax+B) + (Cx+D)e^x + \frac{x^4}{12} + \frac{2}{3}x^3 + 3x^2 + 8x + 10 + \frac{x^2 e^x}{2}$$

Type 4 :

$$\text{If } Q(x) = e^{ax} V(x)$$

Where  $V(x)$  is  $\sin ax$  (or)  $\cos ax$  (or)  $x^n$

Then replace D by D+a for  $e^{ax}$  after that the remaining part will be either procedure like in Type II & Type III

$$P.I = \frac{1}{f(D)} \cdot e^{ax} V(x)$$

$$= e^{ax} \frac{1}{f(D+a)} \cdot V(x) \quad \text{Replace D by D+a}$$

Now  $\frac{1}{f(D+a)} V(x)$  will be evaluated by Type II (or)

Type III

1) Solve  $(D^2 + 4D + 4)y = e^{-2x} \sin x$

Ans:

$$(D^2 + 4D + 4)y = e^{-2x} \cdot \sin x$$

The A.E is

$$(m^2 + 4m + 4) = 0$$

$$(m+2)(m+2) = 0$$

$$m = -2, -2$$

$\therefore$  The roots are real & equal

$$C.F = (Ax + B)e^{mx}$$

$$C.F = (Ax + B)e^{-2x}$$

$$P.I = \frac{1}{f(D)} \cdot A(x)$$

$$= \frac{1}{D^2 + 4D + 4} \cdot e^{-2x} \cdot \sin x$$

$$= e^{-2x} \cdot \frac{1}{(D+2)^2} \cdot \sin x$$

Replace  $D = D + a$

$$a = -2$$

$$D = D - 2$$

$$= e^{-2x} \cdot \frac{1}{(D-2+2)^2} \cdot \sin x$$

$$= e^{-2x} \cdot \frac{1}{D^2} \cdot \sin x$$

Replace  $D^2 = -a^2$

$$D^2 = -(1)^2$$

$$D^2 = -1$$

$$= e^{-2x} \cdot \frac{1}{-1} \cdot \sin x$$

$$P.I = -e^{-2x} \cdot \sin x$$

$$y = C.F + P.I$$

$$y = (Ax + B)e^{-2x} - e^{-2x} \cdot \sin x$$

2) Solve  $(D^2 + 2D + 1)y = (x + e^x)^2$

Ans:

$$(D^2 + 2D + 1)y = (x + e^x)^2$$

$$(D^2 + 2D + 1)y = x^2 + 2xe^x + e^{2x}$$

The A.E is

$$m^2 + 2m + 1 = 0$$

$$(m+1)(m+1) = 0$$

$$m = -1, -1$$

$\therefore$  The roots are real & equal

$$C.F = (A + Bx)e^{mx}$$

$$C.F = (A + Bx)e^{-x}$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 + 2D + 1} \cdot x^2$$

$$= \frac{1}{(D+1)^2} \cdot x^2$$

$$= (D+1)^{-2} \cdot x^2 = (1+D)^{-2} \cdot x^2$$

$$[\because (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots]$$

$$= [1 - 2D + 3D^2 - 4D^3 + \dots] \cdot x^2$$

$$= [x^2 - 2D(x^2) + 3D^2(x^2) - 4D^3(x^2) + \dots]$$

$$= [x^2 - 2(2x) + 3(2) - 4(0) + \dots]$$

$$P.I_1 = x^2 - 4x + 6$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$\frac{1}{D^2 + 2D + 1} \cdot 2x e^x$$

$$= 2e^x \cdot \frac{1}{(D+1)^2} \cdot x$$

$$\text{Rep } D = D+a$$

$$a = 1$$

$$= 2e^x \cdot \frac{1}{(D+2)^2} \cdot x$$

$$= 2e^x \cdot \frac{1}{D^2 + 4D + 4} \cdot x$$

$$= 2e^x \cdot \frac{1}{4 \left[ 1 + \frac{D^2 + 4D}{4} \right]} \cdot x$$

$$= \frac{1}{2} e^x \cdot \left[ 1 + \frac{D^2 + 4D}{4} \right]^{-1} \cdot x$$

$$= \frac{1}{2} e^x \left[ 1 - \left( \frac{D^2 + 4D}{4} \right) + \left( \frac{D^2 + 4D}{4} \right)^2 - \dots \right] \cdot x$$

$$= \frac{1}{2} e^x \left[ 1 - \frac{D^2}{4} - \frac{4D}{4} + \frac{D^4}{16} + \frac{16D^2}{16} + \frac{2(4D^3)}{16} - \dots \right] \cdot x$$

$$= \frac{1}{2} e^x \left[ x - \frac{D^2(x)}{4} - D(x) + \frac{D^4(x)}{16} + \frac{8D^3(x)}{16} + \frac{16D^2(x)}{16} - \dots \right]$$

$$= \frac{1}{2} e^x [x - 0 - 1 + 0 + 0 + 0]$$

$$= \frac{1}{2} e^x (x-1)$$

$$P.I_2 = \frac{e^x (x-1)}{2}$$

$$P.I_3 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 + 2D + 1} \cdot e^{2x}$$

$$\text{Rep } D = a$$

$$a = 2$$

$$= \frac{1}{2^2 + 2(2) + 1} \cdot e^{2x}$$

$$= \frac{1}{4+4+1} \cdot e^{2x}$$

$$P.I_3 = \frac{e^{2x}}{9}$$

$$y = C.F + P.I$$

$$y = (A+Bx)e^{-x} + x^2 - 4x + 6 + \frac{e^x(x-1)}{2} + \frac{e^{2x}}{9}$$

3. Solve  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^x \cos 2x$

Ans:

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^x \cos 2x$$

$$D^2y - 5Dy + 6y = e^x \cos 2x$$

$$(D^2 - 5D + 6)y = e^x \cos 2x$$

The A.E is

$$m^2 - 5m + 6 = 0$$

$$(m-3)(m-2) = 0$$

$$m = 3, 2$$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1x} + Be^{m_2x}$$

$$C.F = Ae^{3x} + Be^{2x}$$

$$P.I = \frac{1}{f(D)} \cdot Q(x)$$

$$P.I = \frac{1}{D^2 - 5D + 6} \cdot e^x \cos 2x$$

$$= e^x \cdot \frac{1}{D^2 - 5D + 6} \cdot \cos 2x$$

$$\begin{aligned} D &= D + a \\ &= D + 1 \end{aligned}$$

$$= e^x \cdot \frac{1}{(D+1)^2 - 5(D+1) + 6} \cdot \cos 2x$$

$$= e^x \cdot \frac{1}{D^2 + 2D + 1 - 5D - 5 + 6} \cdot \cos 2x$$

$$= e^x \cdot \frac{1}{D^2 - 3D + 2} \cdot \cos 2x$$

$$\begin{aligned} \text{Rep } D^2 &= -(2^2) \\ &= -(2^2) \\ &= -4 \end{aligned}$$

$$= e^x \cdot \frac{1}{-4 - 3D + 2} \cdot \cos 2x$$

$$= e^x \cdot \frac{1}{-3D - 2} \cdot \cos 2x$$

$$= -e^x \cdot \frac{1}{3D + 2} \cdot \cos 2x$$

$$= -e^x \cdot \frac{1}{3D + 2} \times \frac{3D - 2}{3D - 2} \cdot \cos 2x$$

$$= -e^x \cdot \frac{3D - 2}{(3D)^2 - 2^2} \cdot \cos 2x$$

$$= -e^x \cdot \frac{(3D - 2) \cos 2x}{9D^2 - 4}$$

$$\begin{aligned} \text{Rep } D^2 &= -a^2 \\ &= -(2^2) \\ &= -4 \end{aligned}$$

$$= -e^x \cdot \frac{3D \cos 2x - 2 \cos 2x}{9(-4) - 4}$$

$$= -e^x \cdot \frac{3D \cos 2x - 2 \cos 2x}{-40}$$

$$= \frac{e^x}{40} \cdot [3(-2 \sin 2x) - 2 \cos 2x]$$

$$= -2 \frac{e^x}{40} [3 \sin 2x + \cos 2x]$$

$$P.I = -\frac{e^x}{20} [3 \sin 2x + \cos 2x]$$

$$y = C.F + P.I$$

$$y = Ae^{3x} + Be^{2x} - \frac{e^x}{20} [3 \sin 2x + \cos 2x]$$

4) Solve  $(D^2 - 2D + 2)y = e^x x^2 + 5 + e^{-2x}$

6m:

$$(D^2 - 2D + 2)y = e^x x^2 + 5 + e^{-2x}$$

The A.E is

$$m^2 - 2m + 2 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1, b=-2, c=2$$

$$m = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$m = \frac{2 \pm \sqrt{-4}}{2}$$

$$m = \frac{2 \pm i\sqrt{4}}{2}$$

$$= 1 \pm \frac{2i}{2}$$

$$m = 1 \pm i$$

$\therefore$  The roots are real & imaginary

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$C.F = e^x (A \cos x + B \sin x)$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 2D + 2} \cdot e^x x^2$$

$$\text{Rep } D = D + a \\ = D + 1$$

$$= e^x \cdot \frac{1}{(D+1)^2 - 2(D+1) + 2} \cdot x^2$$

$$= e^x \cdot \frac{1}{D^2 + 2D + 1 - 2D - 2 + 2} \cdot x^2$$

$$= e^x \cdot \frac{1}{D^2+1} \cdot x^2$$

$$= e^x \cdot (1+D^2)^{-1} \cdot x^2$$

$$\left[ \therefore (1+x^2)^{-1} = 1 - x + x^2 - x^3 + \dots \therefore \right] x^2$$

$$= e^x \cdot [1 - D^2 + (D^2)^2 - (D^2)^3 + \dots] x^2$$

$$= e^x \cdot [1 - D^2 + D^4 - D^6 + \dots] x^2$$

$$= e^x [x^2 - D^2(x^2) + D^4(x^2) - D^6(x^2) + \dots]$$

$$P.I_1 = e^x (x^2 - 2)$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 2D + 2} + 5e^{0x}$$

$$D = a$$

$$D = 0$$

$$= 5 \cdot \frac{1}{0 - 2(0) + 2}$$

$$P.I_2 = 5/2$$

$$P.I_3 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 2D + 2} \cdot e^{-2x}$$

$$D = a$$

$$D = -2$$

$$= \frac{1}{(-2)^2 - 2(-2) + 2} \cdot e^{-2x}$$

$$= \frac{1}{4 + 4 + 2} \cdot e^{-2x}$$

$$P.I_3 = \frac{e^{-2x}}{10}$$

$$Y = C.F + P.I$$

$$Y = e^x (A \cos x + B \sin x) + e^x (x^2 - 2) + \frac{5}{2} + \frac{e^{-2x}}{10}$$

Type - V

1) Solve  $(D^2 - 1)y = x \sin x$

Ans:

$$(D^2 - 1)y = x \sin x$$

$$\left[ \begin{array}{c} e^{i0} = \cos 0 + i \sin 0 \\ \uparrow \quad \quad \downarrow \\ \text{R.P.} \quad \quad \text{I.P.} \end{array} \right]$$

The A.E is

$$m^2 - 1 = 0$$

$$m^2 = 1$$

$$m = \pm 1$$

$$\left[ \begin{array}{l} \cos 0 = \text{R.P. of } e^{i0} \\ \sin 0 = \text{I.P. of } e^{i0} \end{array} \right]$$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^{-x} + Be^x$$

$$P.I = \frac{1}{D^2 - 1} \cdot x \sin x$$

$$[\because e^{iax} = \cos ax + i \sin ax]$$

$$\text{I.P. of } e^{iax} = \sin ax$$

$$= \text{I.P. of } \frac{1}{D^2 - 1} \cdot x \cdot e^{ix}$$

$$= \text{I.P. of } e^{ix} \cdot \frac{1}{(D+i)^2 - 1} \cdot x$$

$$\text{Rep } D = D + a$$

$$a = i$$

$$D = D + i$$

$$= \text{I.P. of } e^{ix} \cdot \frac{1}{D^2 + 2iD + i^2 - 1} \cdot x$$

$$= \text{I.P. of } e^{ix} \cdot \frac{1}{D^2 + 2iD - 2} \cdot x$$

$$= \text{I.P. of } e^{ix} \cdot \frac{1}{-2 \left[ \frac{D^2 + 2iD + 1}{-2} + 1 \right]} \cdot x$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \frac{1}{1 - \left[ \frac{D^2}{2} - iD \right]} \cdot x$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \frac{1}{1 - \left[ iD + \frac{D^2}{2} \right]} \cdot x$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \cdot \left[ 1 - \left[ iD + \frac{D^2}{2} \right] \right]^{-1} \cdot x$$

$$\left[ \because (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots \right]$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \cdot \left[ 1 + \left[ iD + \frac{D^2}{2} \right] + \dots \right] x$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \cdot \left[ x + iD(x) + \frac{D^2}{2}(x) + \dots \right]$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} \cdot [x + i(1) + 0]$$

$$= -\frac{1}{2} \text{I.P of } e^{ix} [x + i]$$

$$= -\frac{1}{2} \text{I.P of } [\cos x + i \sin x] \cdot (x + i)$$

$$= -\frac{1}{2} \text{I.P of } (x \cos x + i \cos x + i x \sin x + i^2 \sin x)$$

$$= -\frac{1}{2} \text{I.P of } (x \cos x + i \cos x + i x \sin x - \sin x)$$

$$= -\frac{1}{2} (\cos x + x \sin x)$$

$$P.I = -\frac{1}{2} (\cos x + x \sin x)$$

$$y = C.F + P.I$$

$$y = Ae^{-x} + Be^x - \frac{1}{2} (\cos x + x \sin x)$$

1) Solve  $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos 2x$

Ans

$$(D^3 - 3D^2 + 4D - 2)y = e^x + \cos 2x$$

The A.E is

$$m^3 - 3m^2 + 4m - 2 = 0$$

$$\begin{array}{r|rrrr} 1 & 1 & -3 & 4 & -2 \\ & 0 & 1 & -2 & 2 \\ \hline & 1 & -2 & 2 & 0 \end{array}$$

The root  $m = 1$

$$m^2 - 2m + 2 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1, b = -2, c = 2$$

$$m = \frac{2 \pm \sqrt{4 - 4(1)(2)}}{2}$$

$$m = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$m = 1 \pm \frac{2i}{2}$$

$$m = 1 \pm i$$

$$C.F = Ae^{m_1 x} + e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$C.F = Ae^x + e^x (A \cos x + B \sin x)$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^3 - 3D^2 + 4D - 2} \cdot e^x$$

$$\text{Rep } D = a$$

$$= \frac{1}{1^3 - 3(1)^2 + 4(1) - 2} \cdot e^x$$

$$= \frac{1}{1 - 3 + 4 - 2} \cdot e^x$$

$$= \frac{1}{0} \cdot e^x$$

$$P.I_1 = x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{3D^2 - 6D + 4} \cdot e^x$$

$$= x \cdot \frac{1}{3(1) - 6(1) + 4} \cdot e^x$$

$$= x \cdot \frac{1}{1} e^x$$

$$P.I_1 = x e^x$$

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^3 - 3D^2 + 4D - 2} \cdot \cos x$$

$$\text{Rep } D^2 = -(a^2) \\ = -1$$

$$= \frac{1}{D^3 + 3 + 4D - 2} \cdot \cos x$$

$$= \frac{1}{D^2 \cdot D + 4D + 1} \cdot \cos x$$

$$= \frac{1}{D+4D+1} \cdot \cos x$$

$$= \frac{1}{3D+1} \cdot \cos x$$

$$= \frac{3D-1}{(3D)^2-1} \cdot \cos x$$

$$= \frac{(3D-1) \cos x}{9D^2-1}$$

$$= \frac{(3D-1) \cos x}{9(-1)-1}$$

$$= \frac{3D \cos x - \cos x}{-10}$$

$$= \frac{(3(-\sin x) - \cos x)}{-10}$$

$$= - \frac{(3 \sin x + \cos x)}{-10}$$

$$P.I. = \frac{3 \sin x + \cos x}{10}$$

$$y = C.F. + P.I.$$

$$y = Ae^x + e^x (A \cos x + B \sin x) + x e^x + \left( \frac{3 \sin x + \cos x}{10} \right)$$

2. Solve  $\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$

Ans

$$\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$$

$$D^3 y + 3D^2 y + 3Dy + 2y = 0$$

$$(D^3 + 3D^2 + 3D + 2)y = 0$$

The A.E is

$$m^3 + 3m^2 + 3m + 2 = 0$$

$$\begin{array}{r|rrrr} -2 & 1 & 3 & 3 & 2 \\ & 0 & -2 & -2 & -2 \\ \hline & 1 & 1 & 1 & 0 \end{array}$$

$$\boxed{m = -2}$$

$$m^2 + m + 1 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1^2 - 4(1)(1)}}{2a}$$

$$= \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{3}i}{2}$$

$$m = -\frac{1}{2} \pm \frac{\sqrt{3}i}{2}$$

$$C.F = Ae^{m_1 x} + e^{\alpha x} (B \cos \beta x + C \sin \beta x)$$

$$C.F = Ae^{-2x} + e^{-\frac{1}{2}x} (B \cos \frac{\sqrt{3}}{2} x + C \sin \frac{\sqrt{3}}{2} x)$$

3. Solve  $(D^2 - 3D + 2)y = 2 \cos(2x + 3) + 2e^x$

Soln:

$$(D^2 - 3D + 2)y = 2 \cos(2x + 3) + 2e^x$$

The A.E is

$$m^2 - 3m + 2 = 0$$

$$(m-2)(m-1) = 0$$

$$\boxed{m=1, 2}$$

∴ The roots are real & distinct

$$C.F = Ae^{m_1 x} + Be^{m_2 x}$$

$$C.F = Ae^x + Be^{2x}$$

$$P.I_1 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 3D + 2} \cdot 2 \cos(2x + 3)$$

$$\begin{aligned} \text{Rep } D^2 &= -a^2 \\ &= -(2)^2 \\ &= -4 \end{aligned}$$

$$= \frac{1}{-4 - 3D + 2} \cdot 2 \cos(2x + 3)$$

$$= \frac{1}{-3D - 2} \cdot 2 \cos(2x + 3)$$

$$= \frac{1}{-(3D + 2)} \cdot 2 \cos(2x + 3)$$

$$= \frac{(3D - 2)}{-(3D)^2 - (2)^2} \cdot 2 \cos(2x + 3)$$

$$= \frac{(3D - 2) \cdot 2 \cos(2x + 3)}{-(9(-1) - 4)}$$

$$= \frac{3D(2 \cos(2x + 3)) - 2(2 \cos(2x + 3))}{40}$$

$$= 3(-2 \sin(2x+3)) \cdot 2 - (4 \cos(2x+3))$$

40

$$= 6(-2 \sin(2x+3)) - (4 \cos(2x+3))$$

40

$$= -2 \cdot 3(2 \sin(2x+3)) + (4 \cos(2x+3))$$

40 20

$$= -6 \sin(2x+3) + 2 \cos(2x+3)$$

20

$$= -2 \frac{3 \sin(2x+3) + \cos(2x+3)}{10}$$

20 10

$$= -3 \sin(2x+3) + \cos(2x+3)$$

10

$$P.I_2 = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 3D + 2} \cdot 2e^x$$

$$= 2 \cdot \frac{1}{1^2 - 3(1) + 2} \cdot e^x$$

$$= 2 \cdot \frac{1}{0} \cdot e^x$$

$$= x \cdot \frac{1}{f'(D)} \cdot Q(x)$$

$$= x \cdot \frac{1}{2D - 3} \cdot 2e^x$$

$$= x \cdot 2 \cdot \frac{1}{2 - 3} \cdot e^x$$

$$= 2x - \frac{1}{1} \cdot e^x$$

$$= -2xe^x$$

$$y = C_1 + C_2 \cdot I$$

$$y = Ae^x + Be^{2x} - \left[ \frac{3 \sin(2x+3) + \cos(2x+3)}{10} \right] \cdot e^{2x}$$

Euler's Homogeneous Linear differential equations:

$$\text{Put } x = e^z$$

$$\Rightarrow \log x = z$$

$$\text{Also } \left[ D = \frac{d}{dz} \right]$$

$$\text{and } xD = \theta$$

$$D = \frac{d}{dz}$$

$$x^2 D^2 = \theta(\theta-1)$$

$$x^3 D^3 = \theta(\theta-1)(\theta-2)$$

⋮

1) Solve  $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = x^2 \cos(\log x)$

Ans:

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = x^2 \cos(\log x)$$

$$x^2 D^2 y - 3x D y + 4y = x^2 \cos(\log x)$$

$$(x^2 D^2 - 3x D + 4)y = x^2 \cos(\log x)$$

Then the transformed eqn is

$$(\theta(\theta-1) - 3\theta + 4)y = e^{2z} \cdot \cos z$$

$$(\theta^2 - \theta - 3\theta + 4)y = e^{2z} \cdot \cos z$$

$$(\theta^2 - 4\theta + 4)y = e^{2z} \cdot \cos z$$

The A.E is

$$m^2 - 4m + 4 = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2$$

∴ The roots are real & equal

$$C.F = (A + Bz)e^{mz}$$

$$C.F = (A + Bz)e^{2z}$$

$$P.I = \frac{1}{\theta^2 - 4\theta + 4} \cdot e^{2z} \cdot \cos z$$

Rep  $\theta = \theta + a$

$$P.I = e^{2z} \cdot \frac{1}{(\theta+2)^2 - 4(\theta+2) + 4} \cdot \cos z \quad a=2$$

$$= e^{2z} \cdot \frac{1}{\theta^2 + 4\theta + 4 - 4\theta - 8 + 4} \cdot \cos z$$

$$= e^{2z} \cdot \frac{1}{\theta^2} \cos z$$

Rep  $\theta^2 = -(a^2)$

$$= -(1)^2$$

$$= -1$$

$$= e^{2z} \cdot \frac{1}{-1} \cos z$$

$$P.I = -e^{2z} \cdot \cos z$$

$$y = C.F + P.I$$

$$y = (A + Bz)e^{2z} - e^{2z} \cdot \cos z$$

Replace  $e^z = x, z = \log x$

$$y = (A + B \log x) x^2 - x^2 \cos(\log x)$$

2) Solve  $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$

Given:

$$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$$

$$x^2 D^2 y - 2x D y - 4y = x^2 + 2 \log x$$

$$(x^2 D^2 - 2x D - 4) y = x^2 + 2 \log x$$

Put  $x = e^z$

$$\log x = z$$

Then  $x D = \theta$

$$x^2 D^2 = \theta(\theta-1)$$

Then the transformed eqn is

$$(\theta(\theta-1) - 2\theta - 4) y = e^{2z} + 2z$$

$$(\theta^2 - \theta - 2\theta - 4) y = e^{2z} + 2z$$

$$(\theta^2 - 3\theta - 4) y = e^{2z} + 2z$$

The A.E is

$$m^2 - 3m - 4 = 0$$

$$(m+1)(m-4) = 0$$

$$m = -1, 4$$

$\therefore$  The roots are real & distinct

$$C.F = A e^{m_1 z} + B e^{m_2 z}$$

$$C.F = A e^{-z} + B e^{4z}$$

$$P.I_1 = \frac{1}{\theta^2 - 3\theta - 4} \cdot e^{2z}$$

$$= \frac{1}{2^2 - 3(2) - 4} \cdot e^{2z}$$

$$\text{Rep } \theta = a$$

$$a = 2$$

$$P.I_1 = -\frac{e^{2z}}{6}$$

$$P.I_2 = 2 \frac{1}{\theta^2 - 3\theta - 4} \cdot z$$

$$P.I_2 = 2 \frac{1}{-4 \left( 1 + \left( \frac{-\theta^2 + 3\theta}{4} \right) \right)} z$$

$$= \frac{1}{-2} \left( 1 + \left( \frac{-\theta^2 + 3\theta}{4} \right) \right)^{-1} \cdot z$$

$$= \frac{1}{-2} \left( 1 - \left( \frac{-\theta^2 + 3\theta}{4} \right) + \left( \frac{-\theta^2 + 3\theta}{4} \right)^2 - \dots \right) \cdot z$$

$$= \frac{1}{-2} \left( 1 - \left( \frac{-\theta^2 + 3\theta}{4} \right) + \left( \frac{\theta^4}{16} + \frac{9\theta^2}{16} - \frac{6\theta^3}{16} \right) - \dots \right) z$$

$$= -\frac{1}{2} \left( z + \frac{\theta^2}{4}(z) - \frac{3\theta}{4}(z) + \frac{\theta^4}{16}(z) + \frac{9\theta^2}{16}(z) - \frac{6\theta^3}{16}(z) - \dots \right)$$

$$= -\frac{1}{2} \left( z + 0 - \frac{3}{4} + 0 + 0 - 0 \right)$$

$$= -\frac{1}{2} \left( z - \frac{3}{4} \right)$$

$$y = C.F + P.I$$

$$y = Ae^{-z} + Be^{4z} - \frac{e^{2z}}{6} - \frac{1}{2} \left( z - \frac{3}{4} \right)$$

$$\text{Rep } e^z = x, z = \log x$$

$$y = Ax^{-1} + Bx^4 - \frac{x^2}{6} - \frac{1}{2} \left( \log x - \frac{3}{4} \right)$$

## Method of Variation of Parameters:

(4) This method is applied for a general second order differential equation of the form

$$y'' + Py' + Qy = X$$

where,  $P, Q$  &  $X$  are functions of  $x$ . The method for finding solution of the above differentials equations as follows

$$y = C.F + P.I$$

Let the complementary function of the given equation is of the form  $C.F = C_1 f_1 + C_2 f_2$  where  $C_1$  &  $C_2$  are constants &  $f_1$  &  $f_2$  are functions of  $x$ .

Then the particular integral is

$$P.I = Pf_1 + Qf_2$$

$$\text{where } P = - \int \frac{f_2 X}{f_1 f_2' - f_1' f_2} dx$$

$$Q = \int \frac{f_1 X}{f_1 f_2' - f_1' f_2} dx$$

$$y = C.F + P.I$$

- 1) Solve  $(D^2 + a^2)y = \tan ax$  by the method of variation of parameters.

$$\text{Gn: } (D^2 + a^2)y = \tan ax$$

The A.E is

$$m^2 + a^2 = 0$$

$$m^2 = -a^2$$

$$m = \pm \sqrt{-a^2} \quad \boxed{m = \pm ai}$$

$$C.F = e^{ax} (C_1 \cos \beta x + C_2 \sin \beta x)$$

$$= e^{ax} (C_1 \cos ax + C_2 \sin ax)$$

$$[ \therefore C.F = C_1 f_1 + C_2 f_2 ]$$

where,

$$f_1 = \cos ax, \quad f_2 = \sin ax$$

$$f_1' = -a \sin ax, \quad f_2' = a \cos ax$$

$$x = \tan ax$$

$$\therefore f_1 f_2' - f_1' f_2 = \cos ax \cdot a \cos ax - [(-a \sin ax) \cdot \sin ax]$$

$$= \cos ax \cdot a \cos ax + a \sin ax \cdot \sin ax$$

$$= a \cos^2 ax + a \sin^2 ax$$

$$= a [\cos^2 ax + \sin^2 ax]$$

To find P.I:

$$= a$$

$$P.I = Pf_1 + Qf_2$$

$$\text{where } P = - \int \frac{f_2 x}{f_1 f_2' - f_1' f_2} dx$$

$$= - \int \frac{\sin ax \cdot \tan ax}{a} \cdot dx$$

$$= - \frac{1}{a} \int \sin ax \cdot \frac{\sin ax}{\cos ax} \cdot dx$$

$$= - \frac{1}{a} \int \frac{\sin^2 ax}{\cos ax} \cdot dx$$

$$= - \frac{1}{a} \int \frac{(1 - \cos^2 ax)}{\cos ax} \cdot dx$$

$$= -\frac{1}{a} \int \left( \frac{1}{\cos ax} \cdot \frac{\cos^2 ax}{\cos ax} \right) dx$$

$$= -\frac{1}{a} \int \left( \frac{1}{\cos ax} \cdot \cos ax \right) dx$$

$$= -\frac{1}{a} \left( \int \sec ax \, dx - \int \cos ax \, dx \right)$$

$$\left[ \because \int \sec ax = \frac{1}{a} \log [\sec ax + \tan ax] \right]$$

$$P = -\frac{1}{a} \left[ \frac{1}{a} \log [\sec ax + \tan ax] - \frac{\sin ax}{a} \right]$$

$$\text{where } Q = \int \frac{f_1 \cdot x}{f_1 f_2' - f_1' f_2} dx$$

$$= \int \frac{\cos ax \cdot \tan ax}{a} dx$$

$$= \frac{1}{a} \int \cos ax \cdot \frac{\sin ax}{\cos ax} dx$$

$$= \frac{1}{a} \int \sin ax \cdot dx$$

$$= \frac{1}{a} \left( -\frac{\cos ax}{a} \right)$$

$$Q = -\frac{\cos ax}{a^2}$$

$$P \cdot I = -\frac{1}{a} \left[ \frac{1}{a} \log [\sec ax + \tan ax] - \frac{\sin ax}{a} \right] \cos ax$$

$$- \frac{\cos ax}{a^2} \cdot \sin ax$$

$$= -\frac{1}{a^2} \log [\sec ax + \tan ax] \cdot \cos ax + \frac{\sin ax \cdot \cos ax}{a^2}$$

$$- \frac{\cos ax \sin ax}{a^2}$$

$$P.I = -\frac{1}{a^2} \log(\sec ax + \tan ax) \cos ax$$

$$y = C.F + P.I$$

$$y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2} \log(\sec ax + \tan ax) \cos ax$$

Using method of variation of parameters

Solve  $\frac{d^2 y}{dx^2} + y = \sec x$

Ans.

$$\frac{d^2 y}{dx^2} + y = \sec x$$

$$D^2 y + y = \sec x$$

$$(D^2 + 1)y = \sec x$$

The A.E is

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m = \pm \sqrt{-1}$$

$$m = \pm i$$

$$C.F = e^{ax} (c_1 \cos px + c_2 \sin px)$$

$$= e^{0x} (c_1 \cos x + c_2 \sin x)$$

$$C.F = c_1 \cos x + c_2 \sin x$$

$$= c_1 f_1 + c_2 f_2$$

$$f_1 = \cos x \quad f_2 = \sin x$$

$$f_1' = -\sin x \quad f_2' = \cos x$$

$$X = \sec x$$

$$\therefore f_1 f_2' - f_1' f_2 = \cos x \cdot \cos x - (-\sin x) \sin x$$

$$= \cos^2 x + \sin^2 x = 1$$

$$P \cdot I = P f_1 + Q f_2$$

$$\text{where } P = - \int \frac{f_2 x}{f_1 f_2' - f_1' f_2} dx$$

$$= - \int \frac{\sin x \cdot \sec x}{1} dx$$

$$= - \int \sin x \cdot \frac{1}{\cos x} dx$$

$$= - \int \tan x dx$$

$$= - \log \sec x$$

$$= - \log \left( \frac{1}{\cos x} \right)$$

$$= - [\log 1 - \log \cos x]$$

$$= - [0 - \log \cos x]$$

$$P = \log \cos x$$

$$Q = \int \frac{f_1 x}{f_1 f_2' - f_1' f_2} dx$$

$$= \int \frac{\cos x \cdot \sec x}{1} dx$$

$$= \int \cos x \cdot \frac{1}{\cos x} dx$$

$$= \int dx$$

$$Q = x$$

$$P.I = P f_1 + Q f_2$$

$$= [\log(\cos x)] \cdot \cos x + x \cdot \sin x$$

$$y = C.F + P.I$$

$$y = C_1 \cos x + C_2 \sin x + [\log(\cos x)] \cdot \cos x + x \cdot \sin x$$

3.  $\frac{d^2 y}{dx^2} + 4y = \tan 2x$

Gn:

$$\frac{d^2 y}{dx^2} + 4y = \tan 2x$$

$$D^2 y + 4y = \tan 2x$$

$$(D^2 + 4)y = \tan 2x$$

The A.E is

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm \sqrt{-4}$$

$$m = \pm 2i$$

$\therefore$  The roots are imaginary

$$C.F = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

$$C.F = C_1 \cos 2x + C_2 \sin 2x$$

$$= C_1 f_1 + C_2 f_2$$

$$f_1 = \cos 2x \quad f_2 = \sin 2x \quad x = \tan 2x$$

$$f_1' = -\sin 2x \cdot 2 \quad f_2' = 2 \cos 2x$$

$$\begin{aligned}
 \therefore f_1 f_2' - f_1' f_2 &= \cos 2x \cdot 2 \cos 2x - (-2 \sin 2x) \cdot \sin 2x \\
 &= 2 \cos^2 2x + 2 \sin^2 2x \\
 &= 2 (\cos^2 2x + \sin^2 2x) \\
 &= 2
 \end{aligned}$$

To find P.I :

$$P.I = P f_1 + Q f_2$$

$$P = - \int \frac{f_2 X}{f_1 f_2' - f_1' f_2} dx$$

$$= - \int \frac{\sin 2x \cdot \tan 2x}{2} dx$$

$$= - \frac{1}{2} \int \sin 2x \cdot \frac{\sin 2x}{\cos 2x} dx$$

$$= - \frac{1}{2} \int \frac{\sin^2 2x}{\cos 2x} dx$$

$$= - \frac{1}{2} \int \frac{(1 - \cos^2 2x)}{\cos 2x} dx$$

$$= - \frac{1}{2} \left( \int \frac{1}{\cos 2x} - \int \frac{\cos^2 2x}{\cos 2x} \right) dx$$

$$= - \frac{1}{2} \left( \int \sec 2x - \int \cos 2x \right) dx$$

$$\left[ \because \sec 2x = \frac{1}{2} \left[ \log [\sec 2x + \tan 2x] \right] \right]$$

$$= - \frac{1}{2} \left[ \frac{1}{2} \log [\sec 2x + \tan 2x] - \frac{\sin 2x}{2} \right]$$

$$= - \frac{1}{4} \log [\sec 2x + \tan 2x] + \frac{\sin 2x}{4}$$

$$Q = \int \frac{f_1 x}{f_1 f_2' - f_1' f_2} dx$$

$$= \int \frac{\cos 2x \cdot \tan 2x}{2} dx$$

$$= \frac{1}{2} \int \cos 2x \cdot \frac{\sin 2x}{\cos 2x} dx$$

$$= \frac{1}{2} \int \sin 2x dx$$

$$= -\frac{1}{2} \frac{\cos 2x}{2}$$

$$= -\frac{\cos 2x}{4}$$

$$P.I = P f_1 + Q f_2$$

$$= -\frac{1}{4} \log [\sec 2x + \tan 2x] \cdot \cos 2x - \frac{\cos 2x}{4} \cdot \sin 2x$$

$$= \left[ -\frac{1}{4} \log [\sec 2x + \tan 2x] + \frac{\sin 2x}{4} \right] \cdot \cos 2x$$

$$- \frac{\cos 2x \cdot \sin 2x}{4}$$

$$= -\frac{1}{4} \log [\sec 2x + \tan 2x] \cos 2x + \frac{\sin 2x \cos 2x}{4}$$

$$- \frac{\cos 2x \sin 2x}{4}$$

$$P.I = -\frac{1}{4} \cos 2x \log [\sec 2x + \tan 2x]$$

$$y = C.F + P.I$$

$$y = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{4} \cos 2x \log$$

$$[\sec 2x + \tan 2x]$$

2) Solve  $(D^2 + 4)y = x^2 \cos 2x$

Ans.

$$(D^2 + 4)y = x^2 \cos 2x$$

The A.E is

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm \sqrt{-4}$$

$$m = \pm 2i$$

$\therefore$  The roots are imaginary.

$$C.F = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

$$C.F = e^{0x} (A \cos 2x + B \sin 2x)$$

$$C.F = (A \cos 2x + B \sin 2x)$$

$$P.I = \frac{1}{f(D)} \cdot Q(x)$$

$$P.I = \frac{1}{D^2 + 4} \cdot x^2 \cos 2x$$

$$= R.P \text{ of } \frac{1}{D^2 + 4} \cdot x^2 \cdot e^{i2x}$$

$$= R.P \text{ of } e^{i2x} \cdot \frac{1}{(D + 2i)^2 + 4} \cdot x^2$$

$$= R.P \text{ of } e^{i2x} \cdot \frac{1}{D^2 + 4iD + 4i^2 + 4} \cdot x^2$$

$$= R.P \text{ of } e^{i2x} \cdot \frac{1}{D^2 + 4iD - 4 + 4} \cdot x^2$$

$$= R.P \text{ of } e^{i2x} \cdot \frac{1}{D^2 + 4iD} \cdot x^2$$

$$\text{Let } D = \sigma + a$$

$$= D + 2i$$

$$\frac{1}{D^2 + 4iD} \cdot x^2$$

$$= R.P. of e^{i2x} \frac{1}{4iD \left[ \frac{D^2}{4iD} + 1 \right]} \cdot x^2$$

$$= R.P. of e^{i2x} \frac{1}{4iD \left[ \frac{D}{4i} + 1 \right]} \cdot x^2$$

$$= R.P. of e^{i2x} \frac{1}{4iD \left[ 1 - \frac{iD}{4} \right]} \cdot x^2$$

$$= R.P. of e^{i2x} \frac{1}{4iD} \left[ 1 - \frac{iD}{4} \right]^{-1} \cdot x^2$$

$$= R.P. of e^{i2x} \frac{1}{4iD} \left[ 1 + \frac{iD}{4} + \left( \frac{iD}{4} \right)^2 + \left( \frac{iD}{4} \right)^3 + \dots \right] x^2$$

$$= R.P. of e^{i2x} \cdot \frac{1}{4i} \left[ \frac{1}{D} + \frac{iD}{4D} - \frac{i^2 D^2}{16D} - \frac{i^3 D^3}{64} + \dots \right] x^2$$

$$= R.P. of e^{i2x} \cdot \frac{1}{4i} \left[ \frac{1}{D} + \frac{i}{4} - \frac{iD}{16} - \frac{i^2 D^2}{64} + \dots \right] x^2$$

$$= R.P. of e^{i2x} \cdot \frac{1}{4i} \left[ \frac{1}{D} (x^2) + \frac{i}{4} (x^2) - \frac{iD}{16} (x^2) - \frac{i^2 D^2}{64} (x^2) + \dots \right]$$

$$= R.P. of e^{i2x} \cdot \frac{1}{4i} \left[ \frac{x^3}{3} + \frac{x^2 i}{4} - \frac{2x i}{16} - \frac{2i}{64} + 0 \right]$$

$$= R.P. of \frac{1}{4i} (\cos 2x + i \sin 2x) \left[ \left( \frac{x^3}{3} - \frac{x}{8} \right) + \frac{i}{4} \left( x^2 - \frac{1}{8} \right) \right]$$

$$= \frac{1}{4} R.P. of -i (\cos 2x + i \sin 2x) \left[ \left( \frac{x^3}{3} - \frac{x}{8} \right) + \frac{i}{4} \left( x^2 - \frac{1}{8} \right) \right]$$

$$= \frac{1}{4} R.P. of (-i \cos 2x + \sin 2x) \left[ \left( \frac{x^3}{3} - \frac{x}{8} \right) + \frac{i}{4} \left( x^2 - \frac{1}{8} \right) \right]$$

$$= \frac{1}{4} R.P. of \left[ \left( \frac{x^3}{3} - \frac{x}{8} \right) (-i \cos 2x) - \frac{i^2}{4} \cos 2x \left( x^2 - \frac{1}{8} \right) + \sin 2x \left( \frac{x^3}{3} - \frac{x}{8} \right) + \frac{i}{4} \left( x^2 - \frac{1}{8} \right) \sin 2x \right]$$

$$P.I = \frac{1}{4} \left[ \frac{\cos 2x}{4} \left( x^2 - \frac{1}{8} \right) + \sin 2x \left( \frac{x^3}{3} - \frac{x}{8} \right) \right]$$

$$y = C.F + P.I$$

$$y = A \cos 2x + B \sin 2x + \frac{1}{4} \left[ \frac{\cos 2x}{4} \left( x^2 - \frac{1}{8} \right) + \sin 2x \left( \frac{x^3}{3} - \frac{x}{8} \right) \right]$$

3) Solve  $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 4 \sin \log x$

Ans:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 4 \sin \log x$$

$$x^2 D^2 y + x Dy + y = 4 \sin \log x$$

$$(x^2 D^2 + x D + 1) y = 4 \sin \log x$$

Put  $x = e^z$

$$\log x = z$$

Then  $x D = \theta$

$$x^2 D^2 = \theta(\theta-1)$$

Replac.

Then the Transformed eqn is

$$(\theta(\theta-1) + \theta + 1) y = 4 \sin z$$

$$(\theta^2 - \theta + \theta + 1) y = 4 \sin z \Rightarrow (\theta^2 + 1) y = 4 \sin z$$

The A.E is

$$m^2 + 1 = 0$$

$$m = \pm \sqrt{-1}$$

$$\boxed{m = \pm i}$$

$\therefore$  The roots are imaginary

$$C.F = e^{\alpha z} (A \cos \beta z + B \sin \beta z)$$

$$= e^{0z} (A \cos z + B \sin z)$$

$$C.F = A \cos z + B \sin z$$

$$P.I = 4 \cdot \frac{1}{\theta^2 + 1} \sin z$$

$$= 4 \cdot \frac{1}{-1+1} \sin z = 4 \cdot \frac{1}{0}$$

$$\text{Rep } \theta^2 = -(a^2)$$

$$= -(1)^2$$

$$= -1$$

$$P.I = 2 \cdot \frac{1}{2\theta} \cdot 4 \sin z$$

$$= 4 \cdot \frac{1}{2} \cdot \frac{1}{\theta} \cdot \sin z$$

$$P.I = -2z \cdot \cos z$$

$$y = C.F + P.I$$

$$y = A \cos z + B \sin z - 2z \cdot \cos z$$

Replace  $z = \log x$

$$y = A \cos \log x + B \sin \log x - 2 \log x \cdot \cos \log x$$

4) Solve  $(x^2 D^2 + 4x D + 2)y = x \log x$

Ans:  $(x^2 D^2 + 4x D + 2)y = x \log x$

$$\text{Put } \Rightarrow x = e^z$$

$$\log x = z$$

$$xD = 0$$

$$x^2 D^2 = \theta(\theta-1)$$

$$(\theta(\theta-1) + 4\theta + 2)y = e^z \cdot z$$

$$(\theta^2 - \theta + 4\theta + 2)y = e^z \cdot z$$

$$(\theta^2 + 3\theta + 2)y = e^z \cdot z$$

The A.E is

$$m^2 + 3m + 2 = 0$$

$$(m+1)(m+2) = 0$$

$$m = -1, -2$$

$\therefore$  The roots are real & distinct

$$C.F = Ae^{m_1 z} + Be^{m_2 z}$$

$$C.F = Ae^{-z} + Be^{-2z}$$

$$P.I = \frac{1}{\theta^2 + 3\theta + 2} \cdot e^z z$$

$$\theta = \theta + a$$

$$= \theta + 1$$

$$P.I = e^z \cdot \frac{1}{(\theta+1)^2 + 3(\theta+1) + 2} \cdot z$$

$$= e^z \cdot \frac{1}{\theta^2 + 2\theta + 1 + 3\theta + 3 + 2} \cdot z$$

$$= e^z \cdot \frac{1}{\theta^2 + 5\theta + 6} \cdot z$$

$$= e^z \cdot \frac{1}{6 \left[ 1 + \frac{\theta^2 + 5\theta}{6} \right]} \cdot z$$

$$= \frac{1}{6} \cdot e^z \cdot \left[ 1 + \frac{\theta^2 + 5\theta}{6} \right]^{-1} \cdot z$$

$$= \frac{1}{6} \cdot e^z \cdot \left[ 1 - \left( \frac{\theta^2 + 5\theta}{6} \right) + \left( \frac{\theta^2 + 5\theta}{6} \right)^2 - \dots \right] z$$

$$= \frac{1}{6} \cdot e^z \left[ 1 - \frac{\theta^2}{6} - \frac{5\theta}{6} + \frac{\theta^4}{36} + \frac{25\theta^2}{36} + \frac{10\theta^3}{36} - \dots \right] z$$

$$= \frac{1}{6} \cdot e^z \left[ z - \frac{\theta^2}{6} (z) - \frac{5\theta}{6} (z) + \frac{\theta^4}{36} (z) + \frac{25\theta^2}{36} (z) + \frac{10\theta^3}{36} (z) - \dots \right]$$

$$= \frac{1}{6} \cdot e^z \left[ z - 0 - \frac{5}{6} + 0 + 0 + 0 \right]$$

$$P.I = \frac{1}{6} \cdot e^z (z - 5/6)$$

$$y = C.F + P.I$$

$$y = Ae^{-z} + Be^{-2z} + \frac{1}{6} \cdot e^z (z - 5/6)$$

Replace  $e^x = x$ ,  $x = \log x$

$$y = Ax^{-1} + Bx^2 + \frac{x}{6} (\log x - \frac{5}{6})$$

Simultaneous Differential Equations with constant coefficients

A pair of equations of the form

$$f_1(D)x + f_2(D)y = \phi_1(t)$$

$$f_3(D)x + f_4(D)y = \phi_2(t), \text{ where } f_1, f_2, f_3 \text{ \& } f_4 \text{ are polynomials in the operator } D = \frac{d}{dt}$$

1) Solve the following simultaneous equations

$$\frac{dx}{dt} + y = \sin t, \quad \frac{dy}{dt} + x = \cos t. \text{ Given that } x=2, y=0 \text{ when } t=0.$$

An eqn can be written as

$$Dx + y = \sin t \quad \text{--- (1)}$$

$$Dy + x = \cos t$$

$$x + Dy = \cos t \quad \text{--- (2)}$$

$$\textcircled{1} \times D \Rightarrow D^2x + Dy = D(\sin t)$$

$$\textcircled{2} \Rightarrow \begin{array}{ccc} x & + & Dy = \cos t \\ (-) & (-) & (-) \end{array}$$

$$(D^2 - 1)x = D(\sin t) - \cos t$$

$$(D^2 - 1)x = \cos t - \cos t$$

$$(D^2 - 1)x = 0$$

$$[\dots x = C \cdot F + P \cdot I]$$

↪  
The A.E is

$$m^2 - 1 = 0$$

$$m^2 = 1$$

$$m = \pm \sqrt{1}$$

$$m = \pm 1$$

∴ The roots are real & distinct

$$C \cdot F = Ae^{m_1 t} + Be^{m_2 t}$$

$$C \cdot F = Ae^{-t} + Be^t \quad P \cdot I = 0$$

$$\boxed{x = Ae^{-t} + Be^t}$$

Diff x w.r.t 't' we get

$$\frac{dx}{dt} = Ae^{-t}(-1) + Be^t$$

$$\frac{dx}{dt} = -Ae^{-t} + Be^t$$

$$\textcircled{1} \Rightarrow Dx + y = \sin t$$

$$\frac{dx}{dt} = \sin t - y$$

$$-Ae^{-t} + Be^t = \sin t - y$$

$$\boxed{y = Ae^{-t} - Be^t + \sin t}$$

∴ The sol's are

$$x = Ae^{-t} + Be^t$$

$$y = Ae^{-t} - Be^t + \sin t$$

Initial conditions are

$$x = 2, y = 0 \text{ when } t = 0$$

$$x = Ae^{-t} + Be^t$$

$$2 = Ae^{-0} + Be^0$$

$$2 = A + B$$

$$\boxed{A + B = 2} \text{ ——— } \textcircled{3}$$

$$y = Ae^{-t} - Be^t + \sin t$$

$$0 = Ae^{(0)} - Be^{(0)} + \sin 0$$

$$\sin 0 = 0$$

$$\cos 0 = 1$$

$$0 = A - B$$

$$\boxed{A = B} \text{ ——— } \textcircled{4}$$

Sub  $\textcircled{4}$  in  $\textcircled{3}$

$$B + B = 2$$

$$2B = 2$$

$$\boxed{B = 1}$$

$$\therefore \boxed{A = 1}$$

$\therefore$  The soln's are

$$x = (e^{-t} + e^t)$$

$$y = e^{-t} - e^t + \sin t$$

Subs A & B values in the soln's of x & y we get

$$x = Ae^t + Be^{-t}$$

$$\text{Subs } A = B = 1$$

$$x = e^t + e^{-t}$$

$$\boxed{x = 2 \cosh t}$$

$$y = Ae^{-t} - Be^t + \sin t$$

$$= e^{-t} - e^t + \sin t$$

$$y = -[e^t - e^{-t}] + \sin t$$

$$\boxed{y = -2 \sinh t + \sin t}$$

2. Solve  $\frac{dx}{dt} - y = t$ ,  $\frac{dy}{dt} + x = t^2$

Given eqns are

$$Dx - y = t \quad \text{--- ①}$$

$$Dy + x = t^2 \quad \text{--- ②}$$

To eliminate  $x$  from ① & ②

$$\text{①} \Rightarrow Dx - y = t$$

$$\text{②} \times D \Rightarrow \underset{(-)}{Dx} + \underset{(-)}{D^2y} = \underset{(-)}{D(t^2)}$$

$$\begin{aligned} -y - D^2y &= t - D(t^2) \\ &= t - 2t \end{aligned}$$

$$-y - D^2y = -t$$

$$\times (-) \Rightarrow y + D^2y = t$$

$$(D^2 + 1)y = t$$

$$y = C.F + P.I$$

The A.E is

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m = \pm \sqrt{-1}$$

$$m = \pm i$$

$\therefore$  The roots are imaginary

$$C.F = e^{\alpha t} (A \cos \beta t + B \sin \beta t)$$

$$= e^{0t} (A \cos t + B \sin t)$$

$$C.F = A \cos t + B \sin t$$

$$P \cdot I = \frac{1}{D^2 + 1} \cdot t$$

$$= [1 + D^2]^{-1} \cdot t$$

$$[ \cdot (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots ]$$

$$P \cdot I = [1 - D^2 + D^4 - D^6 + \dots] \cdot t$$

$$= [t - D^2(t) + D^4(t) - D^6(t) + \dots]$$

$$P \cdot I = t - 0 + 0$$

$$\boxed{P \cdot I = t}$$

$$y = C \cdot F + P \cdot I$$

$$\boxed{y = A \cos t + B \sin t + t}$$

Diff y w.r.t 't'

$$\frac{dy}{dt} = -A \sin t + B \cos t + 1$$

$$\text{From (2)} \Rightarrow Dy + x = t^2$$

$$x = t^2 - Dy \Rightarrow x = t^2 - \frac{dy}{dt}$$

$$x = t^2 - A \sin t + B \cos t + 1$$

$$\boxed{x = t^2 - [A \sin t + B \cos t + 1]}$$

$$x = t^2 + A \sin t - B \cos t - 1$$

∴ The soln's are

$$x = t^2 + c \sin t - D \cos t - 1$$

$$y = A \cos t + B \sin t + t$$

where  $A = c$  &  $B = D$

3) Solve  $\frac{dx}{dt} + 2y = -\sin t$ ,  $\frac{dy}{dt} - 2x = \cos t$

Given eqn

$$Dx + 2y = -\sin t \quad \text{--- ①}$$

$$Dy - 2x = \cos t \quad \text{--- ②}$$

$$\text{①} \times -2 \Rightarrow -2Dx - 4y = 2\sin t$$

$$\text{②} \times D \Rightarrow \begin{matrix} -2Dx & + & D^2y & = & D(\cos t) \\ (+) & & (-) & & (-) \end{matrix}$$

$$-4y - D^2y = -D(\cos t) + 2\sin t$$

$$-4y - D^2y = \sin t + 2\sin t$$

$$-(4y + D^2y) = 3\sin t$$

$$(D^2 + 4)y = -3\sin t$$

The A.E is

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm\sqrt{-4}$$

$$m = \pm 2i$$

$\therefore$  The roots are imaginary

$$C.F = e^{\alpha t} (A \cos \beta t + B \sin \beta t)$$

$$C.F = 0 (A \cos 2t + B \sin 2t)$$

$$C.F = A \cos 2t + B \sin 2t$$

$$P.I = \frac{1}{D^2 + 4} \cdot -3\sin t$$

$$= -3 \frac{1}{-1 + 4} \cdot \sin t$$

$$= -3 \frac{1}{3} \cdot \sin t$$

$$D^2 = -(a^2)$$

$$= -(1^2)$$

$$= -1$$

$$P.I = -\sin t$$

$$y = A \cos 2t + B \sin 2t - \sin t$$

Diff y w.r.t 't'

$$\frac{dy}{dt} = -A \overset{\sin}{\cancel{\cos}} 2t \cdot 2 + B \overset{\cos}{\cancel{\sin}} 2t \cdot 2 - \cos t$$

$$Dy - 2x = \cos t$$

$$\frac{dy}{dt} - 2x = \cos t$$

$$-2x = \cos t - (-A \sin 2t \cdot 2 + B \cos 2t \cdot 2 - \cos t)$$

$$-2x = \cos t + A \sin 2t \cdot 2 - B \cos 2t \cdot 2 + \cos t$$

$$x = A \sin 2t - B \cos 2t + \cos t$$

## 5.7 Short Questions and Answers

**Example 5.7.1.** Solve  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$ .

Jan.'11

**Solution .** Given differential equation is written as,

$$(D^2 - 5D + 6)y = 0.$$

Auxiliary equation (A.E) is,  $m^2 - 5m + 6 = 0$ .

$$\text{i.e., } (m - 2)(m - 3) = 0. \Rightarrow m = 2, 3.$$

Thus the Complementary function (C.F) is  $y = Ae^{2x} + Be^{3x}$ .

**Question 5.7.2.** Solve :  $(D^2 + 3D + 2)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,  $m^2 + 3m + 2 = 0$ .

$$\text{i.e., } (m + 1)(m + 2) = 0. \Rightarrow m = -1, -2.$$

Hence the solution is,  $y = Ae^{-x} + Be^{-2x}$ .

**Question 5.7.3.** Solve :  $(D^2 - 4D + 3)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,  $m^2 - 4m + 3 = 0$ .

$$\text{i.e., } (m - 1)(m - 3) = 0. \Rightarrow m = 1, 3.$$

Thus the solution is,  $y = Ae^x + Be^{3x}$ .

**Question 5.7.4.** Solve :  $(D^2 - 4D + 4)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,  $m^2 - 4m + 4 = 0$ .

$$\text{i.e., } (m - 2)^2 = 0. \Rightarrow m = 2, 2.$$

Thus the solution is,  $y = (A + Bx)e^{2x}$ .

**Question 5.7.5.** Solve :  $(D^2 - 4D + 5)y = 0$ .

**Solution .** Auxiliary equation (A.E) is  $m^2 - 4m + 5 = 0$ .

$$\therefore m = \frac{4 \pm \sqrt{16 - 20}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i.$$

Thus the solution is,  $y = e^{2x}[A \cos x + B \sin x]$ .

**Question 5.7.6.** Solve :  $(D^2 + D + 1)y = 0$ .

May'15

**Solution .** Auxiliary equation (A.E) is  $m^2 + m + 1 = 0$ .

$$\therefore m = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}.$$

Thus the solution is,  $y = e^{x/2} \left[ A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x \right]$ .

**Question 5.7.7.** Solve the differential equation  $(D^2 + 4)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,  $m^2 + 4 = 0$ .

$$\Rightarrow m^2 = -4 = 4i^2. \therefore m = \pm 2i. (\alpha \pm i\beta)$$

Thus the solution is,  $y = A \cos 2x + B \sin 2x$ .

**Question 5.7.8.** Solve :  $(D^3 - 1)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,  $m^3 - 1 = 0$ . Where  $m = 1$  is a factor then by using synthetic division method we get  $m^2 + m + 1 = 0$ .

That is  $(m - 1)(m^2 + m + 1) = 0 \Rightarrow m = 1, \frac{-1 \pm i\sqrt{3}}{2}$ .

Thus the solution is,  $y = Ae^x + e^{x/2} \left[ B \cos \frac{\sqrt{3}}{2}x + C \sin \frac{\sqrt{3}}{2}x \right]$ .

**Question 5.7.9.** Solve :  $(D^4 - 16)y = 0$ .

**Solution .** Auxiliary equation (A.E) is,

$$m^4 - 16 = 0 \Rightarrow k^2 - 16 = 0 \Rightarrow k = \pm 4. \text{ Where } k = m^2.$$

$$\therefore m^2 = \pm 4. \Rightarrow m^2 = 4 \text{ and } m^2 = -4 = 4i^2. \therefore m = \pm 2, \pm 2i.$$

$\therefore$  The roots are  $m = 2, -2, \pm 2i$ .

Thus the solution is,  $y = Ae^{2x} + Be^{-2x} + C \cos 2x + D \sin 2x$ .

**Example 5.7.10.** Solve  $(D^4 + 8D^2 + 16)y = 1$ .

Jan.'11

**Solution .** Auxiliary equation (A.E) is ,

$$m^4 + 8m^2 + 16 = 0 \text{ i.e., } (m^2 + 4)^2 = 0. \Rightarrow m^2 = -4 = 4i^2.$$

$\therefore m = \pm 2i$  twice.  $\therefore$  The roots are  $m = 2i, 2i, -2i, -2i$ .

Hence Complementary function (C.F.) is,

$$y = (A + Bx) \cos 2x + (C + Dx) \sin 2x.$$

To find P.I.:

$$\text{P.I.} = \frac{1}{D^4 + 8D^2 + 16} e^{0x} \quad [\because 1 = e^{0x}]$$

Replace  $D = 0$  we get

$$= \frac{1}{16}.$$

Thus the solution is  $y = \text{C.F.} + \text{P.I.}$

$$= (A + Bx) \cos 2x + (C + Dx) \sin 2x + \frac{1}{16}.$$

Question 5.7.11. Solve :  $(D^2 + 5D + 6)y = e^x$ .

Solution . Auxiliary equation (A.E) is,  $m^2 + 5m + 6 = 0$ .

$$\text{i.e., } (m + 2)(m + 3) = 0. \Rightarrow m = -2, -3.$$

$\therefore$  The complementary function (C.F) is  $y = Ae^{-2x} + Be^{-3x}$ .

To find P.I.:

$$\text{P.I.} = \frac{1}{D^2 + 5D + 6} e^x \quad \text{Replace } D = 1 \text{ we get}$$

$$= \frac{e^x}{1 + 5 + 6} = \frac{e^x}{12}$$

Thus the solution is,

$$y = \text{C.F} + \text{P.I.} = Ae^{-2x} + Be^{-3x} + \frac{e^x}{12}.$$

Question 5.7.12. Find the P.I. of  $(D^2 + 5D + 6)y = e^{-2x}$ .

Solution .

$$\text{P.I.} = \frac{1}{D^2 + 5D + 6} e^{-2x}$$

$$\text{Replace } D = -2 \text{ we get } \frac{e^{-2x}}{4 - 10 + 6} = \frac{e^{-2x}}{0}$$

$$= \frac{x}{2D + 5} e^{-2x} \quad \text{Replace } D = -2$$

$$= \frac{x e^{-2x}}{-4 + 5} = x e^{-2x}.$$

Question 5.7.13. Find P.I. of  $(D^2 - 6D + 9)y = e^{3x}$ .

Solution .

$$\text{P.I.} = \frac{1}{D^2 - 6D + 9} e^{3x}$$

$$\text{Replace } D = 3 \text{ we get } \frac{e^{3x}}{9 - 18 + 9} = \frac{e^{3x}}{0}$$

$$= \frac{x}{2D - 6} e^{3x} \quad \text{Replace } D = 3 \text{ we get } \frac{e^{3x}}{6 - 6} = \frac{e^{3x}}{0}$$

$$= \frac{x^2 e^{3x}}{2}.$$

Question 5.7.14. Find the particular integral of  $(D^2 + 4)y = e^{4x} + 5$ .

Solution .

$$\text{P.I.}_1 = \frac{1}{D^2 + 4} e^{4x} \quad \text{Replace } D = 4 \text{ we get}$$

$$= \frac{e^{4x}}{4^2 + 4} = \frac{e^{4x}}{20}.$$

$$\text{P.I.}_2 = \frac{1}{D^2 + 4} 5e^{0x} \quad \text{Replace } D = 0 \text{ we get}$$

$$= 5 \frac{e^{0x}}{4} = \frac{5}{4}.$$

Example 5.7.15. Find the particular integral of  $(D + 1)^2 y = e^{-x}$ .

Apr.11

Solution .

$$\text{P.I.} = \frac{1}{(D + 1)^2} e^{-x} \quad \text{Replace } D = -1 \text{ we get } \frac{1}{0} e^{-x}$$

$$= \frac{x}{2(D + 1)} e^{-x} \quad \text{Replace } D = -1 \text{ we get } \frac{x}{0} e^{-x}$$

$$= \frac{x^2 e^{-x}}{2}.$$

Example 5.7.16. Find the C.F. of  $(D^2 - 3D + 2)y = e^{5x} + 2$ .

Jan.10

## Higher Order Differential Equations

**Solution .** Given differential equation is  $(D^2 - 3D + 2)y = e^{5x} + 2$ .

Auxiliary equation (A.E.) is ,

$$m^2 - 3m + 2 = 0 \quad \text{i.e., } (m - 1)(m - 2) = 0. \Rightarrow m = 1, 2.$$

Hence Complementary function (C.F) is  $y = Ae^x + Be^{2x}$ .

**Question 5.7.17.** Find P.I. of  $(D^4 - 2D^3 + D^2)y = e^x$ .

**Solution .**

$$\text{P.I.} = \frac{1}{D^4 - 2D^3 + D^2} e^x$$

$$\text{Replace } D = 1 \text{ we get } \frac{e^x}{1 - 2 + 1} = \frac{e^x}{0}$$

$$= \frac{x}{4D^3 - 6D^2 + 2D} e^x$$

$$\text{Replace } D = 1 \text{ we get } \frac{e^x}{4 - 6 + 2} = \frac{e^x}{0}$$

$$= x \frac{x}{12D^2 - 12D + 2} e^x \quad \text{Replace } D = 1 \text{ we get}$$

$$= \frac{x^2 e^x}{2}.$$

**Example 5.7.18.** Find the C.F. of  $(D^2 - 3D + 2)y = e^{5x} + 2$ .

Jan.'10

**Solution .** Given differential equation is  $(D^2 - 3D + 2)y = e^{5x} + 2$ .

Auxiliary equation (A.E.) is ,

$$m^2 - 3m + 2 = 0 \quad \text{i.e., } (m - 1)(m - 2) = 0. \Rightarrow m = 1, 2.$$

Hence Complementary function (C.F) is  $y = Ae^x + Be^{2x}$ .

**Example 5.7.19.** Find the P.I. for  $(D^2 - 3D + 4)y = e^{4x}$ .

Jan.'10

**Solution .**

$$\text{P.I.} = \frac{1}{D^2 - 3D + 4} e^{4x} \quad \text{Replace } D = 4 \text{ we get}$$

$$= \frac{e^{4x}}{16 - 12 + 4} = \frac{e^{4x}}{8}.$$

**Example 5.7.20.** Find the particular integral of  $(D + 1)^2 y = e^{-x}$ .

Apr.11

Solution .

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{(D+1)^2} e^{-x} \quad \text{Replace } D = -1 \text{ we get } \frac{1}{0} e^{-x} \\
 &= \frac{x}{2(D+1)} e^{-x} \quad \text{Replace } D = -1 \text{ we get } \frac{x}{0} e^{-x} \\
 &= \frac{x^2 e^{-x}}{2}.
 \end{aligned}$$

Question 5.7.21. Find particular integral of  $(D^2 + 4)y = \cosh 2x$ .

Solution . Given differential equation is

$$(D^2 + 4)y = \cosh 2x = \frac{1}{2} [e^{2x} + e^{-2x}] = \frac{1}{2} e^{2x} + \frac{1}{2} e^{-2x}.$$

To find P.I.<sub>1</sub>:

$$\begin{aligned}
 \text{P.I.}_1 &= \frac{1}{2} \frac{1}{D^2 + 4} e^{2x} \\
 &\quad \text{Replace } D = 2 \text{ we get} \\
 &= \frac{1}{2} \frac{e^{2x}}{4 + 4} = \frac{e^{2x}}{16}.
 \end{aligned}$$

To find P.I.<sub>2</sub>:

$$\begin{aligned}
 \text{P.I.}_2 &= \frac{1}{2} \frac{1}{D^2 + 4} e^{-2x} \\
 &\quad \text{Replace } D = 2 \text{ we get} \\
 &= \frac{1}{2} \frac{e^{-2x}}{4 + 4} = \frac{e^{-2x}}{16}.
 \end{aligned}$$

Hence the Particular integral solution is

$$\begin{aligned}
 y &= \text{P.I.}_1 + \text{P.I.}_2 \\
 &= \frac{e^{2x}}{16} + \frac{e^{-2x}}{16} \\
 &= \frac{1}{8} \left[ \frac{e^{2x} + e^{-2x}}{2} \right] = \frac{1}{8} \cosh 2x.
 \end{aligned}$$

Question 5.7.22. Find the particular integral of  $(D^2 - 9)y = \sinh 3x$ .

**Solution .** Given differential equation is

$$(D^2 - 9)y = \sinh 3x = \frac{1}{2} [e^{3x} - e^{-3x}] = \frac{1}{2} e^{3x} - \frac{1}{2} e^{-3x}.$$

**To find P.I.<sub>1</sub>:**

$$\begin{aligned} \text{P.I.}_1 &= \frac{1}{2} \frac{1}{D^2 - 9} e^{3x} \quad \text{Replace } D = 3 \text{ we get } \frac{e^{3x}}{0} \\ &= \frac{1}{2} \frac{x}{2D} e^{3x} \quad \text{Replace } D = 3 \text{ we get} \\ &= \frac{x e^{3x}}{12}. \end{aligned}$$

**To find P.I.<sub>2</sub>:**

$$\begin{aligned} \text{P.I.}_2 &= -\frac{1}{2} \frac{1}{D^2 - 9} e^{-3x} \quad \text{Replace } D = -3 \text{ we get } \frac{e^{-3x}}{0} \\ &= -\frac{1}{2} \frac{x}{2D} e^{-3x} \quad \text{Replace } D = -3 \text{ we get} \\ &= -\frac{x e^{-3x}}{4 \cdot -3} = \frac{x e^{-3x}}{12}. \end{aligned}$$

Hence the particular integral solution is

$$\begin{aligned} y &= \text{P.I.}_1 + \text{P.I.}_2 = \frac{x e^{3x}}{12} + \frac{x e^{-3x}}{12} \\ &= \frac{x}{6} \left[ \frac{e^{3x} + e^{-3x}}{2} \right] = \frac{x}{6} \cosh 3x. \end{aligned}$$

**Example 5.7.23.** Find the C.F. of  $(D^2 - 4D + 4)y = e^{2x} + \cos 2x$ . **Apr.12**

**Solution .** Given differential equation is,

$$(D^2 - 4D + 4)y = e^{2x} + \cos 2x.$$

Auxiliary equation (A.E.) is ,

$$m^2 - 4m + 4 = 0 \quad \text{i.e., } (m - 2)^2 = 0. \Rightarrow m = 2, 2.$$

Hence Complementary function (C.F.) is  $y = [A + Bx] e^{2x}$ .

**Question 5.7.24.** Solve :  $(D^2 + 1)y = \sin x$ . **Apr.'14, Jan.'15**

**Solution .** Auxiliary equation (A.E) is,  $m^2 + 1 = 0$ .  $\therefore m = \pm i$ .  
Hence Complementary function (C.F) is,  $y = A \cos x + B \sin x$ .

**To find P.I.:**

$$\text{P.I.} = \frac{1}{D^2 + 1} \sin x$$

Replace  $D^2 = -1^2 = -1$  we get

$$= \frac{1}{-1 + 1} \sin x = \frac{1}{0} \sin x$$

$$= x \frac{1}{2D} \sin x = -\frac{x \cos x}{2}.$$

Hence the solution is,

$$y = \text{C.F.} + \text{P.I.} = A \cos x + B \sin x - \frac{x \cos x}{2}.$$

**Question 5.7.25.** Find the particular integral of  $\frac{d^2 y}{dx^2} + 4y = \sin 2x$ .

Jan.11, May'15

**Solution .** Given differential equation is written as

$$(D^2 + 4)y = \sin 2x.$$

**To find P.I.:**

$$\text{P.I.} = \frac{1}{D^2 + 2^2} \sin 2x$$

$$\text{Replace } D^2 = -2^2 = -4 \text{ we get } \frac{\sin 2x}{-4 + 4} = \frac{\sin 2x}{0}$$

$$= \frac{x}{2} \frac{1}{D} \sin 2x = \frac{x}{2} \int \sin 2x \, dx$$

$$\text{P.I.} = -\frac{x \cos 2x}{4}.$$

**Question 5.7.26.** Find C.F. of  $(D^2 - 4D - 12)y = \sin x \sin 2x$ .

Nov.11

**Solution .** Auxiliary equation (A.E.) is,  $m^2 - 4m - 12 = 0$ .  
i.e.,  $(m + 2)(m - 6) = 0 \Rightarrow m = -2, 6$ .

Thus the Complementary function (C.F.) is,  $y = Ae^{-2x} + Be^{6x}$ .

Question 5.7.27. Find the C.F. of  $(D^2 + 2D + 1)y = \cos 2x$ .

Apr.10

Solution . Given differential equation is  $(D^2 + 2D + 1)y = \cos 2x$ .  
Auxiliary equation (A.E.) is ,  $m^2 + 2m + 1 = 0$ .

$$\text{i.e., } (m + 1)^2 = 0. \Rightarrow m = -1, -1.$$

Thus the Complementary function (C.F) is  $y = (A + Bx) e^{-x}$ .

Question 5.7.28. Find the P.I. for  $(D^2 + 16)y = \cos 4x$ . Apr.10

Solution .

$$\text{P.I.} = \frac{1}{D^2 + 4^2} \cos 4x$$

$$\text{Replace } D^2 = -4^2 = -16 \text{ we get } \frac{\cos 4x}{-16 + 16} = \frac{\cos 4x}{0}$$

$$= \frac{x}{2D} \cos 4x = \frac{x}{2} \frac{1}{D} \cos 4x$$

$$= \frac{x}{2} \int \cos 4x \, dx$$

$$\text{P.I.} = \frac{x \sin 4x}{8}.$$

Question 5.7.29. Solve :  $(D^2 + 4)y = \cos 2x$ .

Solution . Auxiliary equation (A.E) is,  $m^2 + 4 = 0$ .

$$\Rightarrow m^2 = -4 = i^2 4 = \pm i2.$$

Hence Complementary function (C.F) is

$$y = e^{0x} [A \cos 2x + B \sin 2x] = A \cos 2x + B \sin 2x.$$

To find P.I.:

$$\text{P.I.} = \frac{1}{D^2 + 4} \cos 2x$$

$$\text{Replace } D^2 = -2^2 = -4 \text{ we get } \frac{\cos 2x}{0}$$

$$= \frac{x}{2D} \cos 2x = \frac{x}{2} \int \cos 2x \, dx$$

$$= \frac{x}{2} \frac{\sin 2x}{2} = \frac{x \sin 2x}{4}.$$

Hence the complete solution is

$$y = \text{C.F.} + \text{P.I.} = A \cos 2x + B \sin 2x + \frac{x \sin 2x}{4}.$$

Question 5.7.30. Find particular integral of  $(D^2 + a^2)y = \cos ax$ .

Solution . Given differential equation is  $(D^2 + a^2)y = \cos ax$ .

To find P.I.:

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + a^2} \cos ax \\ &\quad \text{Replace } D^2 = -a^2 \text{ we get } \frac{\cos ax}{0} \\ &= \frac{x}{2D} \cos ax \\ &= \frac{x}{2} \frac{D}{D^2} \cos ax \quad \text{Replace } D^2 = -a^2 \text{ we get} \\ &= \frac{x}{2} \left[ \frac{-a \sin ax}{-a^2} \right] = \frac{x \sin ax}{2a}. \end{aligned}$$

Question 5.7.31. Find particular integral of  $(D^2 + a^2)y = \sin ax$ .

Solution .

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + a^2} \sin ax \quad \text{Replace } D^2 = -a^2 \text{ we get } \frac{\sin ax}{0} \\ &= \frac{x}{2D} \sin ax \\ &= \frac{x}{2} \frac{D}{D^2} \sin ax \quad \text{Replace } D^2 = -a^2 \text{ we get} \\ &= \frac{x}{2} \frac{a \cos ax}{-a^2} = -\frac{x \cos ax}{2a}. \end{aligned}$$

Question 5.7.32. Find particular integral of  $(D^2 + 4D + 4)y = \sin 2x$ .

Solution .

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 4D + 4} \sin 2x \quad \text{Replace } D^2 = -2^2 = -4 \text{ we get} \\ &= \frac{1}{-4D} \sin 2x \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{4} \frac{D}{D^2} \sin 2x \quad \text{Replace } D^2 = -2^2 = -4 \text{ we get} \\
 &= -\frac{1}{4} \frac{2 \cos 2x}{-4} = \frac{\cos 2x}{8}.
 \end{aligned}$$

**Question 5.7.33.** Find the P.I. of  $(D^3 - 4D^2 + 4D)y = \cos 2x$ .

**Solution .**

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{(D^2)D - 4D^2 + 4D} \cos 2x \\
 &\quad \text{Replace } D^2 = -2^2 = -4 \text{ we get} \\
 &= \frac{1}{-4D - 4(-4) + 4D} \sin 2x \\
 &= \frac{1}{16} \sin 2x.
 \end{aligned}$$

**Question 5.7.34.** Find particular integral of  $(D^2 + 4)y = \cos^2 x$ .

**Solution .** Given differential equation is written as,

$$(D^2 + 4)y = \cos^2 x = \frac{1 + \cos 2x}{2} = \frac{1}{2} + \frac{\cos 2x}{2}.$$

**To find P.I.<sub>1</sub>:**

$$\begin{aligned}
 \text{P.I.}_1 &= \frac{1}{2} \frac{1}{D^2 + 4} e^{0x} \quad \text{Replace } D = 0 \text{ we get} \\
 &= \frac{1}{2} \frac{e^{0x}}{0 + 4} = \frac{1}{8}.
 \end{aligned}$$

**To find P.I.<sub>2</sub>:**

$$\begin{aligned}
 \text{P.I.}_2 &= \frac{1}{2} \frac{1}{D^2 + 4} \cos 2x \\
 &\quad \text{Replace } D^2 = -2^2 = -4 \text{ we get } \frac{\cos 2x}{0}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \frac{1}{2D} \cos 2x \\
 &= \frac{1}{4} \frac{D}{D^2} \cos 2x \quad \text{Replace } D^2 = -2^2 = -4 \text{ we get} \\
 &= \frac{1}{4} \frac{-2 \sin 2x}{-4} = \frac{\sin 2x}{8}.
 \end{aligned}$$

**Question 5.7.35.** Find particular integral of  $(D^2 + 5D + 6)y = x$ .

**Solution .** Given differential equation is  $(D^2 + 5D + 6)y = x$ .

To find P.I.:

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{6 + 5D + D^2} x \\
 &= \frac{1}{6[1 + (5D + D^2)/6]} x \\
 &= \frac{1}{6} \left[ 1 + \left( \frac{5D + D^2}{6} \right) \right]^{-1} x \\
 &= \left[ 1 - \left( \frac{5D + D^2}{6} \right) + \left( \frac{5D + D^2}{6} \right)^2 - \dots \right] x \\
 &= \left[ 1 - \frac{5D}{6} - \frac{D^2}{6} + \dots \right] x \\
 &= \left[ x - \frac{5}{6} Dx - \frac{1}{6} D^2 x + \dots \right] \\
 &= \left[ x - \frac{5}{6} \cdot 1 - 0 \right] \\
 &= x - \frac{5}{6}.
 \end{aligned}$$

Hence the particular integral solution is  $y = \text{P.I.} = x - 5/6$ .

**Question 5.7.36.** Find particular integral of  $(D^2 - D)y = x^2$ .

**Solution .** Given differential equation is  $(D^2 - D)y = x^2$ .

To find P.I.:

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{-D + D^2} x^2 = \frac{1}{-D(1 - D)} x^2 \\
 &= \frac{1}{-D} [1 - D]^{-1} x^2 \\
 &= -\frac{1}{D} [1 + D + D^2 + D^3 + D^4 \dots] x^2 \\
 &= -\left[ \frac{1}{D} + 1 + D + D^2 + D^3 \dots \right] x^2
 \end{aligned}$$

$$\begin{aligned}
 &= - \left[ \frac{1}{D} x^2 + x^2 + D x^2 + D^2 x^2 + D^3 x^2 + \dots \right] \\
 &= - \left[ \int x^2 dx + x^2 + 2x + 2 + 0 \right] \\
 &= - \left[ \frac{x^3}{3} + x^2 + 2x + 2 \right]
 \end{aligned}$$

Hence the particular integral solution is,

$$y = \text{P.I.} = - \left[ \frac{x^3}{3} + x^2 + 2x + 2 \right].$$

**Question 5.7.37.** Find particular integral of  $(D^2 + 2D + 1)y = x^2 + 1$ .  
**Solution .**

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{D^2 + 2D + 1} (x^2 + 1) \\
 &= \frac{1}{1 + 2D + D^2} (x^2 + 1) \\
 &= \frac{1}{(1 + D)^2} (x^2 + 1) = (1 + D)^{-2} (x^2 + 1) \\
 &= [1 - 2D + 3D^2 - 4D^3 + \dots] (x^2 + 1) \\
 &= [(x^2 + 1) - 2D(x^2 + 1) + 3D^2(x^2 + 1) - 4D^3(x^2 + 1) + \dots] \\
 &= [(x^2 + 1) - 2(2x) + 3(2)] \\
 &= [x^2 + 1 - 4x + 6] \\
 &= [x^2 - 4x + 7].
 \end{aligned}$$

Hence the particular integral solution is  $y = \text{P.I.} = x^2 - 4x + 7$ .

**Example 5.7.38.** Find the complementary function of

$$(D^3 - D^2 - 6D)y = x^2 + 1.$$

Apr.11

**Solution .** Auxiliary equation (A.E) is,  $m^3 - m^2 - 6m = 0$ .  
 i.e.,  $m(m^2 - m - 6) = 0 \Rightarrow m(m + 2)(m - 3) = 0$ .

$\therefore m = 0, -2, 3$ . Thus the Complementary function (CF) is,

$$y = Ae^{0x} + Be^{-2x} + Ce^{3x} = A + Be^{-2x} + Ce^{3x}.$$

**Example 5.7.39.** Find the particular integral of  $(D^2 + 4)y = x^2$ .  
Nov.11

**Solution .**

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{4 + D^2} x^2 \\
 &= \frac{1}{4 \left[ 1 + \frac{D^2}{4} \right]} x^2 \\
 &= \frac{1}{4} \cdot \left[ 1 + \frac{D^2}{4} \right]^{-1} x^2 \\
 &= \frac{1}{4} \left[ 1 - \frac{D^2}{4} + \frac{D^4}{16} - \dots \right] x^2 \\
 &= \frac{1}{4} \left[ x^2 - \frac{1}{4} D^2 x^2 + \frac{1}{16} D^4 x^2 - \dots \right] \\
 &= \frac{1}{4} \left[ x^2 - \frac{1}{2} \right].
 \end{aligned}$$

**Example 5.7.40.** Find particular integral of  $(D^2 + 1)y = x^4$ .

**Solution .**

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{D^2 + 1} x^4 \\
 &= [1 + D^2]^{-1} x^4 \\
 &= [1 - D^2 + D^4 - D^6 + \dots] x^4 \\
 &= [x^4 - D^2(x^4) + D^4(x^4) - D^6(x^4) + \dots] \\
 &\quad D^2(x^4) = D(4x^3) = 4 \cdot 3 \cdot x^2 = 12x^2 \\
 &\quad D^4(x^4) = D^2 D^2(x^4) = D^2(12x^2) = D(24x) = 24 \\
 &\quad D^6(x^4) = D^2 D^4(x^4) = D^2(24) = D D(24) = 0 \\
 &= [x^4 - 12x^2 + 24 + 0] \\
 &= x^4 - 12x^2 + 24.
 \end{aligned}$$

**Question 5.7.41.** Find particular integral of  $(D + 1)^2 y = e^{-x} \cos x$ .  
Jan. '14

Solution . Given differential equation is  $(D + 1)^2 y = e^{-x} \cos x$ .  
To find P.I.:

$$\text{P.I.} = \frac{1}{(D + 1)^2} \cos x$$

Replace  $D$  by  $D - 1$  we get

$$= e^{-x} \frac{1}{(D - 1 + 1)^2} \cos x$$

$$= e^{-x} \frac{1}{D^2} \cos x$$

$$= e^{-x} \frac{1}{D} \sin x$$

$$= -e^{-x} \cos x.$$

Question 5.7.42. Find the particular integral of  $(D^2 - 4D + 4)y = e^{2x} \sin 3x$ .

Solution . Given differential equation is  $(D^2 - 4D + 4)y = e^{2x} \sin 3x$ .  
To find P.I.:

$$\text{P.I.} = \frac{1}{D^2 - 4D + 4} e^{2x} \sin 3x$$

$$= \frac{1}{(D - 2)^2} e^{2x} \sin 3x$$

Replace  $D$  by  $D + 2$  we get

$$= e^{2x} \frac{1}{[(D + 2) - 2]^2} \sin 3x$$

$$= e^{2x} \frac{1}{D^2} \sin 3x$$

Replace  $D^2 = -3^2 = -9$  we get

$$= e^{2x} \frac{\sin 3x}{-9}$$

$$= -\frac{e^{2x} \sin 3x}{9}.$$

Hence the particular integral solution is

$$y = \text{P.I.} = -\frac{e^{2x} \sin 3x}{9}.$$

**Question 5.7.43.** Write the Cauchy's homogeneous linear equation.

**Answer :** An equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = X$$

Where  $X$  is a function of  $x$ , is called as Cauchy's homogeneous linear equation.

**Question 5.7.44.** Write the transformation that transforming the differential equation of variable coefficients into constant coefficients.

**Answer :** By putting  $x = e^z$  or  $z = \log x$  in the linear differential equations with constant coefficients transform to linear differential equations with variable coefficients.

**Question 5.7.45.** Discuss how does the linear differential equation with variable coefficients transform to linear differential equation with constant coefficients?

**Answer :** Consider Cauchy's homogeneous linear equation for variable coefficients

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = X$$

Where  $X$  is a function of  $x$ . Such equations can be reduced to linear differential equations with constant coefficients, by putting  $x = e^z$  or  $z = \log x$ . Then if

$$x \frac{dy}{dx} = \theta; \quad x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1); \quad x^3 \frac{d^3 y}{dx^3} = \theta(\theta - 1)(\theta - 2); \text{ and so on.}$$

After making above substitutions in Cauchy's linear equation we get the following linear differential equations with constant co-efficients form

$$\frac{d^n y}{dz^n} + b_1 \frac{d^{n-1} y}{dz^{n-1}} + \dots + b_{n-1} \frac{dy}{dz} + b_n y = X$$

**Question 5.7.46.** Solve:  $x^2 y'' - xy' + y = 0$ .

**Solution .** Given differential equation is  $x^2 y'' - xy' + y = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$   
then the transformed equation is

$$[\theta(\theta - 1) - \theta + 1] y = 0$$

$$\text{i.e., } [\theta^2 - 2\theta + 1] y = 0.$$

Auxiliary equation is  $(m - 1)^2 = 0 \Rightarrow m = 1, 1$ .

The complementary function is  $y = (A + Bz)e^z$ .

Replace  $z = \log x$  and  $e^z = x$  then the solution becomes

$$y = (A + B \log x) x.$$

**Question 5.7.47.** Solve:  $x^2 y'' - 3xy' + 4y = 0$ .

**Solution .** Given differential equation is  $x^2 y'' - 3xy' + 4y = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$   
then the transformed equation is

$$[\theta(\theta - 1) - 3\theta + 4] y = 0$$

$$\text{i.e., } [\theta^2 - 4\theta + 4] y = 0$$

Auxiliary equation is  $m^2 - 4m + 4 = 0$ .  $(m - 2)^2 = 0 \Rightarrow m = 2, 2$ .

The complementary function is  $y = (A + Bz)e^{2z}$ .

Replace  $z = \log x$  and  $e^z = x$  then the solution becomes

$$y = (A + B \log x) x^2.$$

**Question 5.7.48.** Solve:  $x^2 y'' - xy' + 8y = 0$ .

**Solution .** Given differential equation is  $x^2 y'' - xy' + 8y = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$   
then the transformed equation is,

$$[\theta(\theta - 1) - \theta + 8] y = 0$$

$$\text{i.e., } [\theta^2 - 2\theta + 8] y = 0.$$

Auxiliary equation is

$$m^2 - 2m + 8 = 0$$

$$\therefore m = \frac{2 \pm \sqrt{4 - 32}}{2} = \frac{2 \pm i2\sqrt{7}}{2} = 1 \pm i\sqrt{7}.$$

The complementary function is  $y = e^z (A \cos \sqrt{7}z + B \sin \sqrt{7}z)$ .

Replace  $z = \log x$  and  $e^z = x$  then the solution becomes

$$y = x[A \cos(\sqrt{7} \log x) + B \sin(\sqrt{7} \log x)].$$

**Question 5.7.49.** Solve:  $(x^2 D^2 - 2xD + 2)y = 0$ .

**Solution .** Given differential equation is  $(x^2 D^2 - 2xD + 2)y = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $xD = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$  then the transformed equation is,

$$[\theta(\theta - 1) - 2\theta + 2]y = 0$$

$$\text{i.e., } (\theta^2 - 3\theta + 2)y = 0.$$

Auxiliary equation is  $m^2 - 3m + 2 = 0$ .

$$\Rightarrow (m - 2)(m - 1) = 0. \Rightarrow m = 1, 2.$$

The complementary function is  $y = Ae^z + Be^{2z}$ .

Replace  $z = \log x$  or  $e^z = x$  then the solution is,  $y = Ax + Bx^2$ .

**Question 5.7.50.** Solve:  $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$ .

**Solution .** Given differential equation is  $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $xD = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$  then the transformed equation is

$$[\theta(\theta - 1) - \theta]y = 0$$

$$\text{i.e., } (\theta^2 - 2\theta)y = 0.$$

Auxiliary equation is  $(m^2 - 2m) = 0 \Rightarrow m(m - 2) = 0. \therefore m = 0, 2$

The complementary function is  $y = Ae^{0z} + Be^{2z} = A + Be^{2z}$ .

Replace  $z = \log x$  and  $e^z = x$  then the solution becomes  $y = A + Bx^2$ .

**Question 5.7.51.** Solve:  $x^2 y'' - 20y = 0$ .

**Solution .** Given differential equation is  $x^2 y'' - 20y = 0$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$   
then the transformed equation is

$$(\theta(\theta - 1) - 20)y = 0$$

$$\text{i.e., } (\theta^2 - \theta - 20)y = 0.$$

Auxiliary equation is  $m^2 - m - 20 = 0$ .

$$\Rightarrow (m - 5)(m + 4) = 0. \therefore m = 5, -4.$$

The complementary function is  $y = Ae^{5z} + Be^{-4z}$ .

Replace  $z = \log x$  and  $e^z = x$  then the solution is,  $y = Ax^5 + Bx^{-4}$ .

**Question 5.7.52.** Transform the equation  $xy'' + y' + 1 = 0$  into a linear equation with constant coefficients.

**Solution .** Given differential equation is  $xy'' + y' + 1 = 0$ .

Multiply by  $x$  on both sides we get  $x^2 y'' + xy' + x = 0$

$$\Rightarrow x^2 y'' + xy' = -x.$$

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$   
then the transformed equation is

$$[\theta(\theta - 1) + \theta]y = -e^z$$

$$\Rightarrow \theta^2 y = -e^z$$

$$\frac{d^2 y}{dz^2} = -e^z.$$

**Question 5.7.53.** Transform the equation  $x^2 y'' - 3xy' + 5y = x^2 \sin(\log x)$  into a linear equation with constant coefficients.

**Solution .** Given equation is  $x^2 y'' - 3xy' + 5y = x^2 \sin(\log x)$ .

Put  $x = e^z \Rightarrow \log x = z$  and  $x D = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$

then the transformed equation is

$$\begin{aligned} [\theta(\theta - 1) - 3\theta + 5] y &= e^{2z} \sin z \\ \Rightarrow [\theta^2 - 4\theta + 5] y &= e^{2z} \sin z. \text{ Where } \theta = \frac{d}{dz} \end{aligned}$$

**Question 5.7.54.** Transform the equation  $(x^2 D^2 + 4xD + 2) y = x \log x$  into a linear equation with constant coefficients.

**Solution .** Given equation is  $(x^2 D^2 + 4xD + 2) y = x \log x$ .

Put  $x = e^z$ .  $\Rightarrow \log x = z$  and  $xD = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$  then the transformed equation is

$$\begin{aligned} [\theta(\theta - 1) + 4\theta + 2] y &= e^z z \\ \Rightarrow (\theta^2 + 3\theta + 2) y &= e^z z. \text{ Where } \theta = \frac{d}{dz} \end{aligned}$$

**Question 5.7.55.** Solve:  $r \frac{d^2 u}{dr^2} - \frac{du}{dr} = 0$ .

**Solution .** Given differential equation is  $r \frac{d^2 u}{dr^2} - \frac{du}{dr} = 0$ .

Multiply by  $r$  on both sides we get  $r^2 \frac{d^2 u}{dr^2} - r \frac{du}{dr} = 0$ .

Put  $r = e^z$ .  $\Rightarrow \log r = z$  and  $xD = \theta$ ,  $x^2 D^2 = \theta(\theta - 1)$  and  $\theta = \frac{d}{dz}$  and  $D = \frac{d}{dr}$  then the transformed equation is

$$\begin{aligned} [\theta(\theta - 1) + \theta] u &= 0 \\ \text{i.e., } \theta^2 u &= 0. \end{aligned}$$

Auxiliary equation is  $m^2 = 0$ .  $\Rightarrow m = 0, 0$ .

The complementary function is  $y = (A + Bz)e^{0z} = A + Bz$ .

Replace  $z = \log r$  and  $e^z = r$  then the solution is,  $y = A + B(\log r)$ .

**Question 5.7.56.** How to do the method of variation of parameters?  
Apr.12

**Answer:** This method is applied when the differential equation is of the form  $y'' + py' + qy = X$ . Where  $p$ ,  $q$  and  $X$  are functions of  $x$ . It

gives the particular integral (P.I.) =  $-y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$ . Where  $y_1$  and  $y_2$  are the solutions of  $y'' + py' + qy = 0$  and  $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$  is called the Wronskian of  $y_1$  and  $y_2$ .

**Proof:**

Let the complementary function of the given equation be  $y = c_1 y_1 + c_2 y_2$ . Replacing  $c_1, c_2$  (regarded as parameters) by unknown functions  $u(x)$  and  $v(x)$ .

Let the particular integral be  $y = u y_1 + v y_2$ .

Differentiate (1) w.r.to  $x$ , we get ...(1)

$$y' = u y_1' + v y_2' + u' y_1 + v' y_2 = u y_1' + v y_2'.$$

On assuming that  $u' y_1 + v' y_2 = 0$ .

Differentiate (1) and substitute in the given equation, then  $y_1$  and  $y_2$  satisfy the equation  $y'' + p y' + q y = 0$ , we obtain ...(2)

$$u' y_1' + v' y_2' = X.$$

Solving equations (2) and (3) we get ...(3)

$$u' = -\frac{y_2 X}{W} \text{ and } v' = \frac{y_1 X}{W}. \quad \text{Where } W = y_1 y_2' - y_2 y_1'.$$

Integrating,  $u = -\int \frac{y_2 X}{W} dx, v = \int \frac{y_1 X}{W} dx$ .

Substituting these in (1) we get  $y = -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$ .

**Question 5.7.57.** Eliminate  $y$  from the simultaneous equations  $\frac{dy}{dt} + y = \sin t, \frac{dx}{dt} + x = \cos t$ .

Jan.'15

**Solution .** Given equations can be written as,

$$Dx + y = \sin t \quad \dots(1)$$

$$Dy + x = \cos t \quad \dots(2)$$

Solving  $y$  from the above two equations,

$$(1) \times D \Rightarrow D^2 x + Dy = D(\sin t)$$

$$\text{and } (2) \Rightarrow Dy - x = \cos t$$

Subtracting the above two equations, we get

$$(D^2 - 1)x = D(\sin t) - \cos t = \cos t - \cos t$$

$$\therefore (D^2 - 1)x = 0.$$